Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Geological Society of America Bulletin

Potassium-Argon Ages and Paleomagnetism of the Waianae and Koolau Volcanic Series, Oahu, Hawaii

RICHARD R. DOELL and G. BRENT DALRYMPLE

Geological Society of America Bulletin 1973;84;1217-1242
doi: 10.1130/0016-7606(1973)84<1217:PAAPOT>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

RICHARD R. DOELL
G. BRENT DALRYMPLE } *U.S. Geological Survey, Menlo Park, California 94025*

Potassium-Argon Ages and Paleomagnetism of the Waianae and Koolau Volcanic Series, Oahu, Hawaii

ABSTRACT

Paleomagnetic and potassium-argon measurements on 786 oriented cores from 99 volcanic units at 18 sites in the Waianae and Koolau Ranges, Oahu, when combined with data from previous studies, show that the subaerial Waianae Volcano was active only from about 3.6 to 2.4 m.y. ago and the subaerial Koolau Volcano from about 2.6 to 1.8 m.y. ago. There is some evidence that Waianae Volcano was still active when Koolau Volcano emerged from the sea. The predominantly tholeiitic lower and middle members of the Waianae Volcanic Series are approximately contemporaneous and were extruded during the late Gilbert and early Gauss geomagnetic polarity epochs. They were followed within less than 0.2 m.y. by the alkalic lavas of the upper member, which were probably extruded largely during the later part of the Gauss normal polarity epoch. The Koolau Volcanic Series was extruded entirely during the early Matuyama reversed polarity epoch. Data from three thick stratigraphic sections in the Waianae and Koolau Volcanic Series indicate that stacks of lava flows as much as 470 m thick can be formed in less than 0.25 m.y. and that the maximum average period between superimposed lava flows is on the order of 10^3 yrs. Additional data on the hawaiite flow that led to the discovery of the Kaena reversed event indicate that this reversed flow is 2.85 ± 0.05 m.y. old. Angular dispersion of virtual geomagnetic poles (VGP) in the Hawaiian Islands appears to have decreased during the past 5 m.y. This may be caused by a decrease in dipole wobble, a decrease in the nondipole component of the Earth's magnetic field, or the accumulated effects of weathering, tectonism, and geomorphic processes in older rocks. The mean Waianae and Koolau VGPs are slightly on the side of the Earth's rotation axis away from Oahu. This supports, but does not prove, the hypothesis that the axial dipole is displaced slightly northward from the Earth's center. Three VGP "excursions" were recorded in sections of lava in the Waianae and Koolau ranges. During these excursions, the VGPs appear to have traveled away from or toward the geographic axis along great circle paths, suggesting they may be related to the dipole rather than the nondipole field and may record aborted reversals in polarity or rapid and infrequent dipole tilts.

INTRODUCTION

This study was conducted to determine the age and duration of eruption of the two shield volcanoes that form the island of Oahu, to obtain data for the time scale of geomagnetic reversals, and to obtain further information about the nature of the Earth's magnetic field in the central Pacific during late Pliocene and early Pleistocene time.

The Hawaiian-Emperor chain of shield volcanoes extends in two roughly linear segments, from the island of Hawaii on the southeast, along the Hawaiian Archipelago and Emperor Seamount Chain, to a point near the western end of the Aleutian Trench on the northwest. A current hypothesis is that this chain was formed by sequential volcanic eruptions as the Pacific plate moved over a melting spot in the mantle, now centered just north of the island of Hawaii (Wilson, 1963; Morgan, 1971, 1972; McDougall, 1971; Jackson and others, 1972). Clearly, the ages of individual Hawaiian volcanoes are of critical importance to hypotheses of the origin of the Hawaiian Islands and of Pacific plate motion during Tertiary time.

Of equal importance is the observation (Cox and Doell, 1964; Doell and Cox, 1965, 1971) that the central Pacific has been an area of unusually low geomagnetic secular variation during the last 0.7 m.y. The Hawaiian Islands

Geological Society of America Bulletin, v. 84, p. 1217–1242, 9 figs., April 1973

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

are an unusually good place to study the nature of the Earth's ancient main dipole field because of this relative lack of nondipole "noise" and because thick sections of well-exposed lava flows exist there. The length of time during which this unusual condition has persisted is also of interest, for it may be an important clue to the nature of the mantle beneath the Pacific.

Oahu, with an area of 1,560 sq km, is the third largest island in the Hawaiian group. It consists primarily of two coalesced shield volcanoes, the Waianae and Koolau, each now deeply eroded. The Koolau Volcano (Fig. 1) is the younger of the two and was active after the Waianae Volcano had ceased to erupt. The geology of Oahu has been described by Stearns and Vaksvik (1935), Stearns (1939, 1940, 1966), and Macdonald and Abbott (1970), from which the following geological summary is taken.

The Waianae Volcanic Series is subdivided into lower, middle, and upper members (Fig. 1). The lower member, which forms the main mass of the volcano, has an exposed thickness of about 600 m and consists of thin flows of tholeiitic lava. The middle member, also about 600 m thick, is formed by lava that filled the Waianae caldera. In most places, it is separated from the lower member by an angular unconformity and by talus breccia that accumulated against the cliffs of the caldera. The lava of the middle member is mostly tholeiitic, but toward the top, flows of alkalic basalt appear. The transition between the middle and upper members is mostly compositional; within the caldera, there is no sharp break between them. The upper member, which once formed a rela-

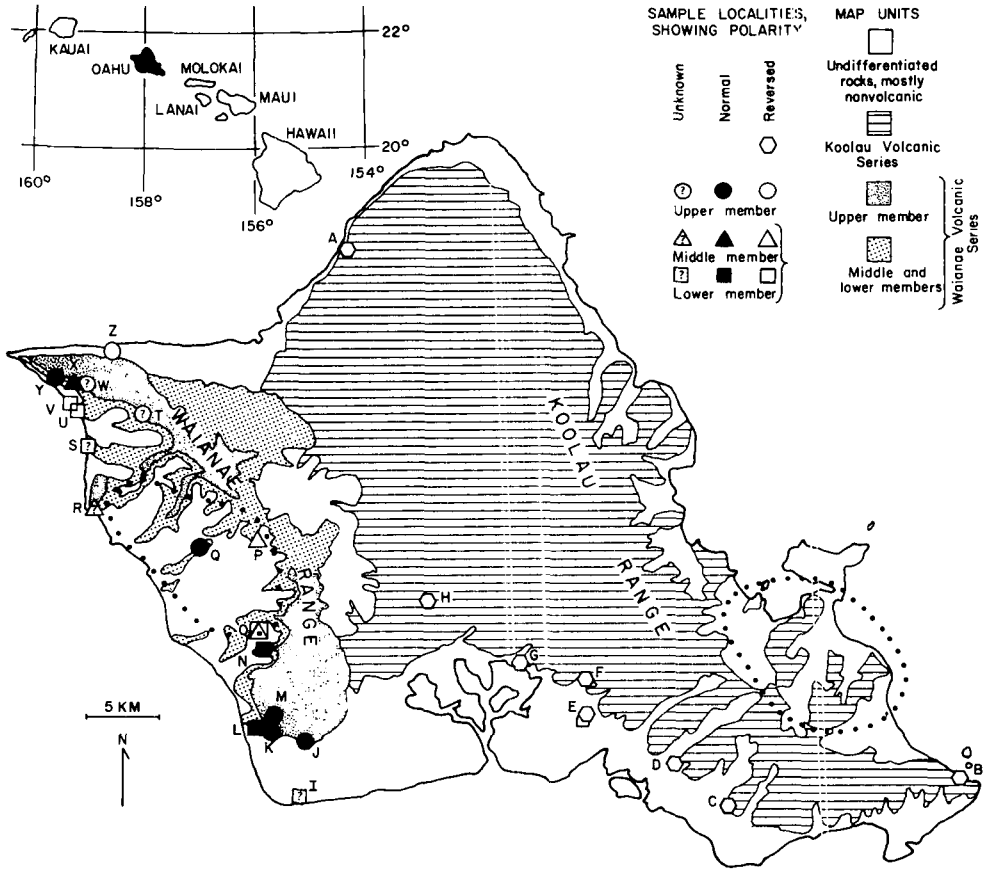


Figure 1. Generalized geologic map of the island of Oahu showing the location of sample sites. Geology after Stearns (1939, 1940). Heavy dotted lines show the approximate locations of the Waianae and Koolau calderas on the basis of gravity data (Strange and others, 1965; Macdonald and Abbott, 1970).

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

tively thin cap covering most of the Waianae Volcano, is about 700 m thick and is composed mostly of massive flows of hawaiite with some alkalic basalt.

The extra-caldera lavas of Koolau Volcano, which consist almost entirely of tholeiitic basalt, are called the Koolau Volcanic Series. Alkalic basalt occurs only as a few late, isolated flows, which indicates that the alkalic stage of Koolau Volcano had barely begun when eruptions ceased. Approximately 900 m of Koolau lava is exposed. After a long period of erosion, during which Koolau Volcano was modified to about its present geomorphic state, 30 to 40 scattered eruptions occurred on the southeastern end of the Koolau Range. The rocks of these localized cones and flows, named the Honolulu Volcanic Series, range in composition from alkalic olivine basalt to nephelinite (Winchell, 1947; Jackson and Wright, 1970). Potassium-argon and paleomagnetic data show that these eruptions occurred within the last 0.7 to 0.8 m.y. (Gramlich and others, 1971; Doell, 1972a; G. B. Dalrymple, unpub. data).

Potassium-argon ages on Hawaiian basalt were first reported by McDougall (1963, 1964), McDougall and Tarling (1963), and McDougall and Chamalaun (1966), who found that the age of the islands increased from Hawaii in the southeast (less than 800,000 yrs) to Kauai on the northwest (5.6 m.y.). On Oahu, McDougall (1964) and McDougall and Ur-Rahman (1972) dated 13 lavas at 13 sites, 8 from the Waianae Range, and 5 from the Koolau Range. The oldest age found was 8.4 m.y. on biotite from the rhyodacite that forms Mauna Kuwale. More recent work by Funkhouser and others (1966) has shown that this rock is more nearly 2 to 3 m.y. old and that the biotite must contain extraneous Ar^{40}. McDougall's potassium-argon ages for other lavas of the Waianae Volcanic Series range from 2.9 to 3.3 m.y. He concluded from these data that the lower and middle members of the Waianae Volcanic Series were essentially contemporaneous and were somewhat older than the upper member, that the change in composition from tholeiitic to alkalic basalt occurred in about 0.2 m.y., and that the Waianae Volcano was active for about 0.7 m.y. In accord with the geologic evidence that the Koolau Volcanic Series is younger than the Waianae Volcanic Series, McDougall found ages ranging from 2.2 to 2.6 m.y. for Koolau lavas. Funkhouser (1966) and Funkhouser and others (1966, 1968) reported potassium-argon ages on 13 lavas from the Waianae Range, and their results are in general agreement with those of McDougall, except for the rhyodacite of Mauna Kuwale discussed above.

Tarling (1962, 1965) and McDougall and Tarling (1963) studied the paleomagnetism of basalt from five Hawaiian Islands including Oahu, on which they found both normally and reversely magnetized rocks. From his data, Tarling (1965) concluded that the mean Hawaiian direction was close to an axial dipole, that secular variation in the Hawaiian Islands may have been normal during the last 5 m.y., and that the lack of nondipole field in the central Pacific observed by Doell and Cox (1965) for Quaternary time may be a short-term phenomenon.

RESULTS

Details of the locations of sampling sites, sampling methods, and analytical techniques are given in Appendix 1. Sample sites are shown on Figure 1.

Paleomagnetism

The mean (flow average) RM direction of each volcanic unit, calculated by Fisher's (1953) method, is given in Tables 1 and 2 for the Waianae and Koolau Volcanic Series, respectively, along with the peak AF strength used for magnetic cleaning, statistical parameters describing the precision of the mean RM directions, and the coordinates of the virtual geomagnetic pole (VGP). The VGP is the calculated pole of a geocentric magnetic dipole that would produce the observed magnetic direction in the absence of all other magnetic field sources (Cox and Doell, 1960). VGP's are more useful than field directions when comparing paleomagnetic data from geographically separated regions.

Not all specimens collected (Tables 1 and 2, col. 2) were used in the flow-average analyses of Tables 1 and 2. The selection of peak strengths for AF cleaning and the rejection of data from flow-average analyses are done to produce the average RM direction for each unit that most probably represents the original TRM acquired by the lava when it cooled. For some flows, the anomalous magnetization, which probably is produced by lightning (Graham, 1961; Cox, 1961), could not be removed from all specimens by AF demagnetization. For these flows, the

TABLE 1. FLOW-AVERAGE PALEOMAGNETIC DATA FOR BASALTS OF THE WAIANAE VOLCANIC SERIES, OAHU

Unit number	Number of samples†	Maximum lateral distance	A.F. cleaning (peak oe)	RM decl. (deg. E)	RM incl. (deg. dn.)	Length of resultant R	Precision parameter k_e	95% confidence α_{95} (deg.)	Angular std. dev. (deg.)	VGP lat (deg. N.)	VGP long (deg. E.)
Middle member, NW. Waianae Range dike (Site X)											
65	8 (8)	24	50	2.5	22.9	7.9909	771	2.0	2.5	80.1	7.4
Middle member, Kolekole Pass (Site P)											
64	8 (8)	41	0	170.8	−25.4	7.9944	1250	1.6	2.3	−78.0	250.6
Upper member, Mauna Kauale Dome (Site Q)											
63	8 (8)	34	50	12.8	53.3	7.9386	114	5.2	7.6	73.2	241.5
Upper member, N. Waianae Range (Site Z)											
62	8 (8)	56	50	171.9	−40.4	7.9412	119	5.1	7.4	−82.4	304.3
Upper member, S.-SW. Waianae Range (Site K)											
61	8 (8)	22	50	353.5	27.7	7.9703	236	3.6	5.3	80.9	65.8
60	8 (8)	13	0	1.0	26.7	7.9680	219	3.8	5.5	82.7	13.9
59	8 (8)	12	50	353.5	29.0	7.9838	433	2.7	3.9	84.1	23.4
Lower member, SW. Waianae Range (Site L)											
58	8 (8)	12	0	339.8	39.2	7.9929	984	1.8	2.6	71.3	118.2
57	8 (8)	15	0	339.7	38.5	7.9890	635	2.2	3.2	71.1	116.6
Upper member, S. Waianae Range (Site J)											
56	9 (9)	37	100	4.6	27.7	8.9761	335	2.8	4.4	82.1	347.6
Lower member, Koluakauila Stream (Site V)											
55	8 (8)	36	100	171.6	−35.5	7.9474	133	4.8	7.0	−81.9	279.7
54	7 (7)	21	0	182.1	−18.8	6.9716	211	4.2	5.6	−77.9	191.8
Middle (43–47) and upper (48–53) members, NW. Waianae Range (43–47 = Site X, 48–53 = Site Y)											
53*	6 (8)	21	400	22.7	54.4	5.7807	23	14.3	17.1	66.0	252.9
52*	8 (8)	15	200	3.3	25.2	7.7653	30	10.3	14.9	81.1	0.5
51	7 (8)	10	400	352.4	29.3	6.8916	55	8.2	10.9	80.7	73.9
50	8 (8)	9	200	342.8	36.7	7.9512	144	4.6	6.8	73.9	111.0
49	8 (8)	19	50	8.6	16.7	7.9831	415	2.7	4.0	74.6	348.0
48	8 (8)	13	100	5.7	16.4	7.9858	492	2.5	3.6	75.7	358.2
47	8 (8)	18	100	5.8	19.7	7.9771	306	3.2	4.6	77.3	354.9
46	8 (8)	27	100	1.8	29.1	7.9810	369	2.9	4.2	83.8	5.7
45	8 (8)	13	50	357.5	32.4	7.9584	168	4.3	6.2	85.4	52.9
44	8 (8)	70	0	356.4	36.8	7.9764	296	3.2	4.7	86.5	95.6
43	8 (8)	22	50	354.2	37.6	7.9869	534	2.4	3.5	84.6	107.5
Lower (14–18) and middle (19–42) members, N. Nanakuli Valley (Site O)											
42	8 (8)	13	200	177.8	−45.5	7.9201	88	6.0	8.7	−84.0	2.6
41	6 (8)	16	200	190.0	−52.0	5.9649	142	5.6	6.8	−75.6	57.8

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

40	8 (8)	12	400	204.3	−47.7	7.8544	48	8.1	11.7	−66.8	88.3
39	7 (8)	22	400	178.5	0.5	6.9630	162	4.8	6.4	−68.3	205.9
38*	8 (8)	24	200	181.6	2.4	7.7926	34	9.7	14.0	−67.4	197.8
37*	7 (8)	7	400	186.2	10.4	6.806	31	11.0	14.6	−62.7	188.3
36	6 (8)	25	200	177.9	−0.5	5.9412	85	7.3	8.8	−68.8	207.8
35	8 (8)	14	50	183.1	3.9	7.9807	362	2.9	4.3	−66.5	194.1
34	8 (8)	23	200	182.8	27.3	7.8314	42	8.7	12.6	−54.0	197.2
33*	8 (8)	24	200	183.7	52.6	7.9130	80	6.2	9.0	−35.4	198.0
32*	5 (8)	18	200	179.6	79.6	4.9145	47	11.3	11.9	1.2	202.0
31*	5 (8)	15	400	161.0	75.2	4.8717	31	13.9	14.6	−5.1	210.6
30*	4 (8)	9	400	163.7	−12.8	3.9157	36	15.6	13.6	−68.0	250.2
29	8 (8)	7	200	187.9	−35.1	7.9513	144	4.6	6.8	−82.3	125.3
28	8 (8)	45	100	182.4	−14.0	7.9902	716	2.1	3.0	−75.6	192.3
27	6 (8)	36	600	182.4	−14.2	5.9623	133	5.8	7.0	−75.7	192.2
26	8 (8)	24	200	184.3	−31.6	7.9587	169	4.3	6.2	−84.1	157.5
25	7 (8)	28	100	179.0	−35.8	6.9964	1676	1.5	2.0	−88.2	232.6
24	6 (8)	23	200	177.0	−21.9	5.9680	156	5.4	6.5	−79.6	218.7
23	7 (8)	16	400	192.3	−25.7	6.9946	1108	1.8	2.4	−75.9	143.4
22	8 (8)	5	100	180.7	−34.0	7.9880	586	2.3	3.4	−87.2	188.1
21	8 (8)	11	200	181.6	−7.9	7.9824	397	2.8	4.1	−72.5	196.7
20	8 (8)	32	200	166.6	−8.5	7.9849	463	2.6	3.8	−68.5	241.1
19	8 (8)	32	100	168.7	−5.2	7.9517	145	4.6	6.7	−68.3	233.8
18	8 (8)	33	100	170.3	−34.7	7.9868	528	2.4	3.5	−80.6	279.7
17	8 (8)	11	200	167.8	−17.5	7.9514	144	4.6	6.8	−73.0	247.1
16	8 (8)	12	200	173.6	−30.7	7.9756	287	3.3	4.8	−82.3	254.4
15	8 (8)	21	200	184.1	−16.3	7.9757	288	3.3	4.8	−76.4	184.3
14	7 (7)	6	0	147.0	−39.9	6.9583	144	5.0	6.8	−59.5	300.7
Lower member, E. Nanakuli Valley (Site N)											
13	7 (8)	36	400	6.4	34.2	6.9815	324	3.4	4.5	83.4	314.6
12	8 (8)	20	600	23.0	32.3	7.9439	125	5.0	7.3	68.0	298.0
11	8 (8)	28	200	20.6	26.5	7.9461	130	4.9	7.1	69.1	309.2
10	8 (8)	20	100	346.3	45.5	7.9505	141	4.7	6.8	76.4	138.6
9	9 (9)	21	100	349.6	48.2	8.9066	86	5.6	8.8	77.8	153.6
8	8 (8)	23	200	341.6	47.2	7.9305	101	5.6	8.1	71.9	138.2
7*	6 (8)	27	400	353.6	51.9	5.9222	64	8.4	10.1	77.5	176.3
6	8 (8)	25	100	358.2	38.8	7.9886	613	2.2	3.3	88.2	128.0
5	9 (9)	26	50	359.3	38.3	8.9852	540	2.2	3.5	89.3	122.4
4	8 (8)	27	0	358.4	35.5	7.9902	717	2.1	3.0	87.6	62.9
3	8 (8)	24	100	356.5	34.3	7.9560	159	4.4	6.4	85.9	74.4
2	8 (8)	14	50	6.6	6.5	7.9665	209	3.8	5.6	70.8	1.6
1*	7 (8)	26	200	37.3	−32.7	6.7898	29	11.5	15.2	36.4	336.1

† Number of samples used in flow average analysis (number of samples collected).
* Not used in paleomagnetic and variance analyses.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 2. FLOW-AVERAGE PALEOMAGNETIC DATA FOR BASALTS OF THE KOOLAU VOLCANIC SERIES, OAHU

Unit number	Number of samples†	Maximum lateral distance	A.F. cleaning (peak oe)	RM decl. (deg. E)	RM incl. (deg. dn.)	Length of resultant R	Precision parameter k_e	95% confidence α_{95} (deg.)	Angular std. dev. (deg.)	VGP lat (degree N.)	VGP long. (degree E.)
Salt Lake (Site E)											
99	8 (8)	35	NRM	166.0	−49.4	7.9224	90	5.9	8.6	−74.6	330.5
Makapuu Head (Site B)											
98	8 (8)	23	200	173.0	−46.9	7.9817	382	2.8	4.2	−80.7	340.9
97	8 (8)	34	100	176.4	−43.3	7.9836	426	2.7	3.9	−84.9	342.9
96	6 (8)	19	400	172.6	−39.1	5.9808	260	4.2	5.0	−83.0	300.4
95	8 (8)	28	100	181.2	−31.2	7.9802	353	3.0	4.3	−85.4	187.8
94	7 (8)	30	200	180.9	−31.9	6.9801	302	3.5	4.7	−85.9	190.7
93	7 (8)	7	400	180.8	−31.9	6.9816	326	3.4	4.5	−85.9	191.0
92	8 (8)	10	200	178.2	−30.2	7.9902	715	2.1	3.0	−84.6	178.2
91	8 (8)	18	200	176.1	−27.2	7.9883	597	2.3	3.3	−82.2	231.2
90	8 (8)	29	200	174.0	−12.0	7.9728	258	3.5	5.0	−73.7	224.0
89	8 (8)	45	100	176.6	−26.1	7.9601	176	4.2	6.1	−81.8	226.0
88	8 (8)	32	0	195.4	−21.0	7.9747	277	3.3	4.9	−71.9	145.2
87	8 (8)	11	0	192.1	−20.6	7.9804	357	2.9	4.3	−74.2	152.9
86	8 (8)	10	50	174.4	−17.1	7.9835	424	2.7	3.9	−76.3	226.5
85	8 (8)	11	50	171.5	−17.9	7.9765	298	3.2	4.7	−75.4	237.7
84	8 (8)	5	50	179.8	−40.6	7.9839	434	2.7	3.9	−88.1	15.5
83	8 (8)	43	0	179.3	−39.6	7.9610	179	4.2	6.1	−88.7	353.6
82	8 (8)	14	0	184.9	−41.8	7.9752	283	3.3	4.8	−84.7	79.8
81	8 (8)	19	0	192.3	−18.4	7.9802	354	3.0	4.3	−73.2	155.4
80	8 (8)	36	50	188.4	−15.0	7.9862	508	2.5	3.6	−74.1	170.4
79	7 (8)	46	100	187.6	−16.8	6.9950	1189	1.8	2.4	−75.3	171.1
78	8 (8)	16	50	176.5	−5.4	7.9886	613	2.2	3.3	−71.1	213.1
77	7 (8)	13	0	195.7	−19.4	6.9684	190	4.4	5.9	−71.1	146.8
76	8 (8)	9	0	176.5	−43.6	7.9845	453	2.6	3.8	−84.8	345.1
East Loch Pearl Harbor (Site G)											
75	8 (8)	163	100	176.2	−11.6	7.9429	123	5.0	7.3	−74.1	215.8
Waianae Bay (Site A)											
74	8 (8)	16	100	182.6	−19.6	7.9826	402	2.8	4.0	−78.2	189.4
73	7 (8)	17	50	183.6	−46.4	6.9910	667	2.3	3.1	−83.1	50.0
North Halawa Stream (Site F)											
72*	6 (8)	32	0	189.3	−79.4	5.9846	325	3.7	4.5	−41.5	26.4
71	8 (8)	58	0	173.5	−48.6	7.9662	207	3.9	5.6	−79.9	347.8
70	9 (9)	30	100	171.0	−31.1	8.9459	148	4.2	6.7	−80.3	265.4
69	8 (8)	61	200	170.5	−23.5	7.9848	461	2.6	3.8	−77.2	248.4
Univ. of Hawaii (Site C)											
68	8 (8)	24	0	164.4	−45.8	7.9364	110	5.3	7.7	−74.6	318.0
67	8 (8)	37	0	160.7	−46.4	7.9680	219	3.8	5.5	−71.4	316.2
Nuuanu Valley (Site D)											
66	8 (8)	25	50	174.9	−16.8	7.9868	532	2.4	3.5	−76.4	224.1

* Not included in paleomagnetic and variance analyses.

† Number of samples used in analysis (number of samples collected).

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

specimens with unremoved anomalous magnetization were not used in the flow-average analyses. An example, unit 27, is shown in Figure 2. Four of the specimens from this unit had NRM directions with southerly declinations and shallow negative inclinations. The other four specimens had NRM directions that were widely scattered. After AF demagnetization at 600 oersteds, the four well-grouped specimens changed only slightly, and the grouping improved. The RM directions of two of the scattered specimens moved into the group defined by the well-grouped specimens, but anomalous RM remained in the other two specimens even after AF demagnetization at 600 oersteds peak strength. The best estimate of the original TRM direction is obtained by excluding the two divergent directions from the flow-average analysis. Similar reasoning was used for other units where it was obvious that anomalous magnetization remained after AF demagnetization.

The general RM stability of the lavas sampled is summarized in Figure 3 in histograms that show the intensity of RM remaining after partial AF demagnetization. For this illustration, a single specimen with minimum anomalous magnetization was included from each lava flow. The RM typically decreases from about 4×10^{-3} emu/cc before treatment to about one-tenth of this after AF demagnetization at 800 oersteds. This behavior is typical of Hawaiian lava flows (Doell and Cox, 1961, 1963, 1965; Doell, 1969, 1972a, 1972b, 1972c) and indicates that the RM in the Waianae and Koolau lava flows, as given in Tables 1 and 2, is predominantly TRM acquired when the lavas cooled. Therefore, the declinations and inclinations in Tables 1 and 2 are reasonable estimates of the declination and inclination of the geomagnetic field at the time and place of cooling of the individual lavas. The precision with which ancient field directions are estimated by these data is indicated by the statistical parameters in columns 8 to 10, Tables 1 and 2, and the over-all precision at the 95-percent level of confidence summarized in Figure 4. Most of the Waianae lavas and all of the Koolau lavas studied have RM directions that define the ancient field directions within about 6° or better. This is sufficiently accurate for the purposes of this investigation.

Potassium-Argon Dating

The analytical data and calculated potassium-argon ages are given in Table 3. A total of 92 argon measurements were made during this investigation, but many of these were of

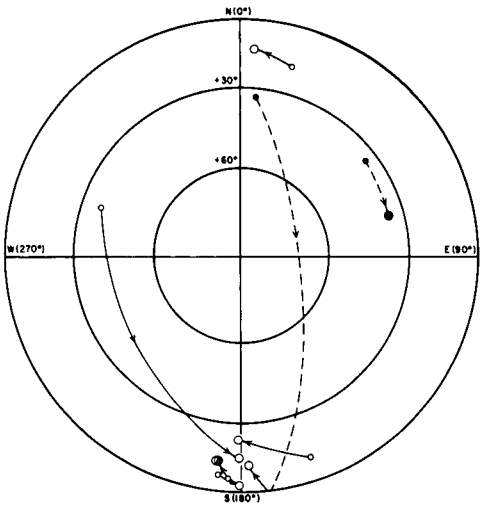


Figure 2. Directions of NRM (small symbols), and RM (large symbols) after demagnetization at 600 oersteds peak field strength, in cores from unit 27. Open symbols and solid lines on upper hemisphere, solid symbols and dashed lines on lower hemisphere of a Lambert equal-area projection.

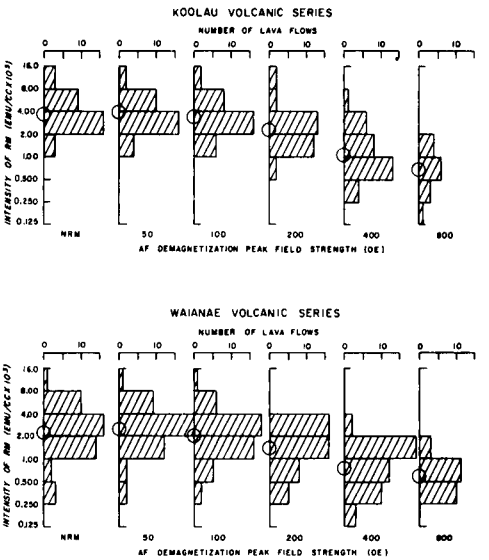


Figure 3. Histograms of the RM remaining in lava of the Waianae and Koolau Volcanic Series after partial AF demagnetization at progressively higher peak field strengths. Circles indicate the mean value for each histogram.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 3A. ANALYTICAL DATA AND POTASSIUM-ARGON AGES OF BASALTS FROM THE WAIANAE VOLCANIC SERIES, OAHU

Site	Unit number*	K_2O (wt percent)	Argon: Weight (grams)	Argon: Ar^{40}_{rad}/gram (10^{-12} moles)	Argon: $10\ Ar^{40}_{rad}/Ar^{40}_{total}$	Calculated age† (10^6 yrs)
N	07 a)	0.336, 0.344	11.206	1.893	12.2	3.76 ± 0.28
	b)	0.366, 0.366	11.544	1.913	12.5	3.53 ± 0.26
	13 a)	0.324, 0.324	8.294	1.790	13.9	4.05 ± 0.17
			10.401	2.090	15.1	
	b)	0.383, 0.385	9.833	1.789	15.4	3.15 ± 0.18
O	15	0.398, 0.402	8.044	1.003	10.4	1.70 ± 0.15
	19	0.778, 0.778	9.609	3.022	42.7	2.63 ± 0.06
			12.251	3.020	61.7	
	20	0.753, 0.755	10.636	3.086	41.6	2.77 ± 0.08
	31	0.681, 0.679	11.128	3.140	17.6	3.12 ± 0.15
	41 a)	0.603, 0.604	12.388	2.743	13.2	2.61 ± 0.16
			10.560	1.920	28.2	
	b)	0.570, 0.569	11.440	2.122	8.4	2.61 ± 0.17
			11.801	2.276	13.6	
Y	43	0.323, 0.330	10.308	1.310	9.3	2.95 ± 0.17
			10.586	1.531	12.8	
	46	0.458, 0.461	8.727	2.305	12.6	3.34 ± 0.17
			10.020	2.228	12.7	
	48 a)	0.373, 0.374	11.321	1.756	12.1	3.29 ± 0.22
			8.470	1.881	8.7	
	b)	0.382, 0.380	7.978	1.901	10.4	3.38 ± 0.30
	52 a)	0.780, 0.783	9.896	2.979	57.7	2.68 ± 0.06
			8.493	3.222	30.8	
	b)	0.806, 0.807	6.648	3.623	11.0	2.90 ± 0.12
			10.552	3.278	44.0	
V	54 a)	0.388, 0.384	8.039	1.653	12.1	3.00 ± 0.16
			10.690	1.770	12.1	
	b)	0.359, 0.359	9.5050	1.494	16.4	2.55 ± 0.14
	55	0.313, 0.312	12.446	1.330	9.7	3.12 ± 0.23
			9.821	1.554	8.4	
J	56	1.220, 1.220	9.958	5.943	26.1	3.30 ± 0.10
K	59	0.905, 0.914	9.821	4.342	16.5	3.10 ± 0.10
			7.971	3.997	20.6	
	61	0.923, 0.919	9.028	3.877	23.6	3.06 ± 0.08
			8.535	4.451	21.0	
Z	62 a)	0.764, 0.760	11.710	3.276	33.2	2.93 ± 0.06
			12.726	3.331	32.6	
	b)	0.792, 0.800	12.981	3.248	16.4	2.90 ± 0.08
			11.208	3.587	31.2	
	c)	0.914, 0.925	12.356	3.674	10.2	2.70 ± 0.24

* Letters a, b, and c designate different cores from the same flow.

† $\lambda_\epsilon = 0.585 \times 10^{-10}\ yr^{-1}$, $\lambda_\beta = 4.72 \times 10^{-10}\ yr^{-1}$, $K^{40}/K_{total} = 1.19 \times 10^{-4}$ mol/mol. The ± figures are estimates of the standard deviation of precision (Cox and Dalrymple, 1967).

marginal quality because of high atmospheric argon contamination and, since better measurements at the same or nearby localities were available, contributed little or no useful additional information to the study. Accordingly, argon measurements with radiogenic Ar^{40} less than 8 percent of the total Ar^{40} were rejected, except for data from the section near Makapuu Head (site B, units 76, 78, 97, and 98) in the Koolau Range. At this site, nine of the ten measurements showed less than 8 percent radiogenic Ar^{40}. Because there are so few data in this part of the Koolau Range, all of the Makapuu data have been included in the table. Of 92 argon measurements, 28 percent (13 Waianae data and 14 Koolau data) have been excluded, leaving a total of 65 argon and 78 K_2O measurements on 39 cores from 27 units.

A standard deviation of precision for each calculated age has been estimated using the method of Cox and Dalrymple (1967). These estimates are usually in reasonable agreement with actual precision parameters found from replication experiments (Cox and Dalrymple, 1967; Dalrymple and Lanphere, 1969). This is true even for samples (including basalts) with

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 3B. ANALYTICAL DATA AND POTASSIUM-ARGON AGES OF BASALTS FROM THE KOOLAU VOLCANIC SERIES, OAHU

Site	Unit number*	K_2O (wt percent)	Argon: Weight (grams)	Argon: Ar^{40}_{rad}/gram (10^{-12} moles)	Argon: $\frac{100\ Ar^{40}_{rad}}{Ar^{40}_{total}}$	Calculated age† (10^6 yrs)
D	66 a)	0.408, 0.408	10.585	1.754	9.7	2.60 ± 0.17
			11.186	1.380	11.0	
	b)	0.416, 0.418	11.438	1.595	10.5	2.05 ± 0.18
			9.726	0.936	9.8	
C	67	0.339, 0.356	11.509	1.323	8.9	2.56 ± 0.18
			10.814	1.306	9.6	
F	69	0.358, 0.363	10.795	0.993	12.2	1.86 ± 0.14
	72	0.420, 0.423	11.476	1.250	9.7	1.86 ± 0.16
			11.625	1.072	11.6	
A	74 a)	0.583, 0.584	12.021	2.435	8.9	2.32 ± 0.17
			9.367	1.567	11.3	
	b)	0.518, 0.522	11.426	1.043	17.9	1.36 ± 0.06
G	75 a)	0.447, 0.458	10.488	1.311	10.0	1.96 ± 0.18
	b)	0.640, 0.612	10.588	0.810	15.2	1.41 ± 0.07
			11.819	1.794	14.2	
B	76 a)	0.109, 0.112	9.581	0.291	3.0	1.75 ± 0.42
			9.251	0.280	2.8	
	b)	0.095, 0.100	9.453	0.372	3.2	2.26 ± 0.45
			10.042	0.279	4.7	
	78	0.335, 0.337	9.404	1.179	14.9	1.59 ± 0.13
			13.008	0.396	5.0	
	97	0.140, 0.138	8.597	0.295	5.1	2.30 ± 0.47
			9.530	0.602	4.0	
	98	0.131, 0.135	10.610	0.386	7.1	1.80 ± 0.31
			11.926	0.320	4.7	

* Letters a, b, and c designate different cores from the same flow.

† $\lambda_\epsilon = 0.585 \times 10^{-10}\ yr^{-1}$, $\lambda_\beta = 4.72 \times 10^{-10}\ yr^{-1}$, $K^{40}/K_{total} = 1.19 \times 10^{-4}$ mol/mol. The ± figures are estimates of the standard deviation of precision (Cox and Dalrymple, 1967).

low percentages and very small quantities of radiogenic Ar^{40} (see, for example, Dalrymple, 1968, 1971). For the Oahu data, however, the precision actually achieved is not so good as expected. If, for statistical uniformity, only samples with more than 0.3 percent of K_2O are considered, then 10 units have potassium-argon ages on more than one core. The pooled estimate (Crow and others, 1960) of the standard deviation, s_0, of these replicate ages is 16 percent, compared with an average (found by arithmetically averaging variances) of 6 percent for the estimated standard deviations obtained by the method of Cox and Dalrymple (1967). The lack of precision in the ages is due almost entirely to imprecision in the argon analyses. Duplicate argon measurements were made on 22 cores with $K_2O > 0.3$ percent, and s_0 for these is 23 percent compared with only 1 percent for 35 pairs of K_2O measurements. The cause of this decline in precision of the argon measurements and hence of the calculated ages must be due to some property of the rocks, perhaps small-scale inhomogeneities in argon distribution within the samples. Such imprecision did not occur with other samples either before, during, or since the Oahu analyses were made and appears to be unique to the Waianae and Koolau Volcanic Series samples. Similar problems of reproducibility are apparent in the data of Funkhouser (1966) and Funkhouser and others

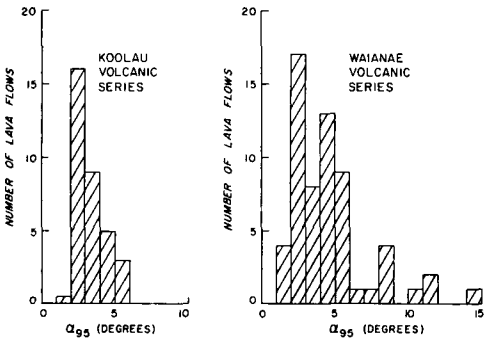


Figure 4. Histograms of 95-percent-confidence cone semiangles (α_{95}) from RM directions measured in lavas of the Waianae and Koolau Volcanic Series.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

(1968) but rarely in the data of McDougall (1964) and McDougall and Ur-Rahman (1972). Since the techniques used by McDougall are essentially the same as those used in this study, the question is presently unanswered.

DISCUSSION

Age of the Waianae and Koolau Volcanic Series

All previously published age data on the Waianae and Koolau Volcanic Series and the results of this study are summarized by site in Figure 5. The only data not shown in the figure are two ages reported by Funkhouser and others (1968) for the lower (B2, 5.4 ± 0.3 m.y.) and upper (B1, 2.8 ± 0.1 m.y.) members of the Waianae Volcanic Series for which locations are not given either in that publication or in Funkhouser (1966).

Over-all, the results are in general agreement and show that the exposed parts of the two main shield volcanoes of Oahu were built within the last 3 to 4 m.y., that the Waianae Volcanic Series is older than the Koolau Volcanic Series, and that Koolau Volcano became

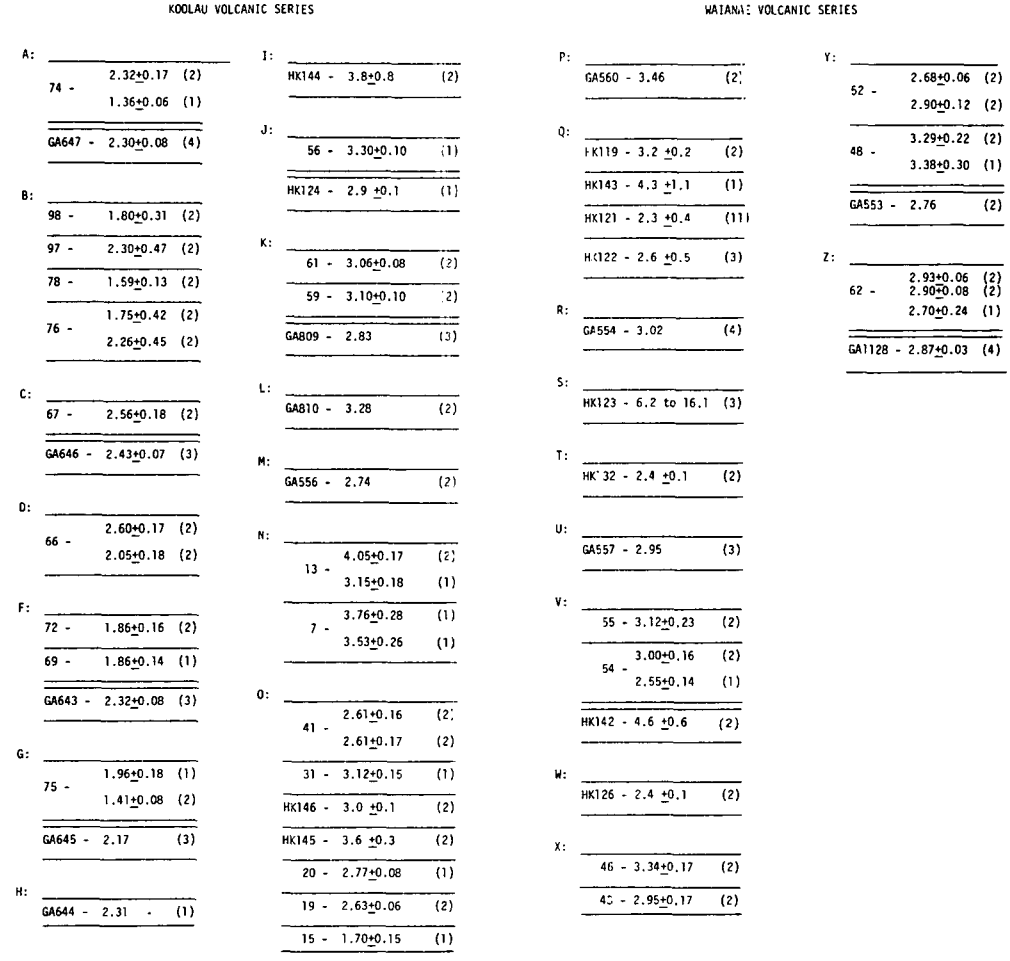


Figure 5. Summary by site (Fig. 1) of all available potassium-argon ages on the Koolau and Waianae Volcanic Series, Oahu, Hawaii. Single horizontal line separates volcanic units whose stratigraphic relations are known (youngest on top); double horizontal line indicates that relation is unknown. Numbers in parentheses indicate the number of potassium-argon measurements on the same sample. Sample numbers prefixed GA are from McDougall (1964), McDougall and Chamalaun (1966), and McDougall and Ur-Rahman (1972); those prefixed HK are from Funkhouser and others (1968); unprefixed numbers are from this study.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

inactive between about 1.5 and 2 m.y. ago. A closer inspection of Figure 5, however, shows that age data from a single site are sometimes inconsistent, probably due to the difficulties with precision discussed above. For this reason, we have selected a "best" age for each site (Table 4) on the assumption that the lava flows at a single locality are unlikely to be of detectably different ages. This assumption is probably justified in view of the rapid rates of extrusion of Hawaiian volcanoes during their shield-building phase (McDougall, 1964; Doell and Cox, 1965; Macdonald and Abbott, 1970; Moore, 1970; Jackson and Wright, 1970; Swanson, 1972). For the same reason, a single magnetic polarity was assigned to each site (Table 4), supported by the fact that rocks of different polarity were never found at the same site (Tables 1 and 2). The "best" ages were selected largely on the basis of reproducibility, stratigraphic consistency, and over-all quality as indicated by the amount of atmospheric Ar^{40} and the estimated precision. The justification for the selection at each site is given in Appendix 2.

The data of Table 4 are plotted in Figure 6 with the mean age and standard deviation for the Koolau Volcanic Series and the upper, middle, and lower members of the Waianae Volcanic Series. In order to allow for data of different quality, these mean ages, shown by shaded crosses in the figure, were calculated by weighting the individual ages of Table 4 by the inverse of their variances. The weighted means in themselves have little geological significance but are useful indicators of the presence or absence of general age differences between mappable units.

TABLE 4. "BEST" POTASSIUM-ARGON AGES AND MAGNETIC POLARITIES FOR THE OAHU SITES SHOWN IN FIGURES 1 AND 5

Locality	Best potassium-argon age* (10^6 yrs)	Polarity†	Formation
A	2.3 ± 0.1	R	Koolau
B	1.8 ± 0.3	R	Koolau
C	2.5 ± 0.1	R	Koolau
D	2.6 ± 0.2	R	Koolau
F	1.9 ± 0.1	R	Koolau
G	2.2 ± 0.1	R	Koolau
H	2.3 ± 0.1	R	Koolau
I	3.8 ± 0.8	U	Waianae, lower member
J	3.1 ± 0.1	N	Waianae, upper member
K	3.0 ± 0.1	N	Waianae, upper member
L	3.3 ± 0.1	N	Waianae, lower member
M	2.7 ± 0.1	N	Waianae, upper member
N	3.6 ± 0.1	N	Waianae, lower member
O	3.0 ± 0.4	R	Waianae, middle member
P	3.5 ± 0.2	R	Waianae, middle member
Q	2.7 ± 0.2	N	Waianae, upper member
R	3.0 ± 0.1	U	Waianae, middle member
T	2.4 ± 0.1	U	Waianae, upper member
U	3.0 ± 0.1	R	Waianae, lower member
V	3.1 ± 0.1	R	Waianae, lower member
W	2.4 ± 0.1	U	Waianae, upper member
X	3.1 ± 0.1	N	Waianae, middle member
Y	3.0 ± 0.3	N	Waianae, upper member
Z	2.8 ± 0.1	R	Waianae, upper member

* The figures are estimated standard deviations of precision. See text.

† N = normal, R = reversed, U = polarity unknown. See Appendix for discussion of individual sites.

At the 90-percent and 95-percent levels of confidence, there is no significant difference in the weighted mean ages of the lower and middle members of the Waianae Volcanic Series. These two units appear to be approximately contemporaneous within the precision of the ages, a conclusion that was reached by McDougall (1964) on the basis of fewer age measurements. This conclusion is also consistent with the geological observation (Stearns and Vaksvik, 1935; Stearns, 1940; Macdonald and Abbott, 1970) that both the lower and middle members, although in a definite stratigraphic sequence, were formed during the main constructional phase of Waianae Volcano. The apparent differences in mean age between the upper member and the middle member and between the upper member and the lower member are real at the 90-percent level of confidence but not at the 95-percent level of confidence. Some additional resolution is provided by the paleomagnetic data. The lower and middle members have both normally and reversely magnetized lavas, and detailed correlation with the polarity time scale is uncertain. The age data indicate, however, that these two members were formed during the late Gilbert and early Gauss polarity epochs. The upper member is predominantly of normal polarity, and the ages clearly indicate that it was formed during the Gauss normal epoch. According to Holmes (1965), who places the Pliocene-Pleistocene boundary at 2 to 3 m.y., the Waianae Volcanic Series is considered Pliocene and Pleistocene(?).

The mean Koolau age is significantly different from the mean ages of all members of the Waianae Volcanic Series at the 95-percent level of confidence. Furthermore, all lavas of the Koolau Volcanic Series sampled by Tarling (1965) and by us have reversed polarity, which, when combined with the age data, strongly indicates that the subaerial Koolau Volcano was formed entirely during the early half of the

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

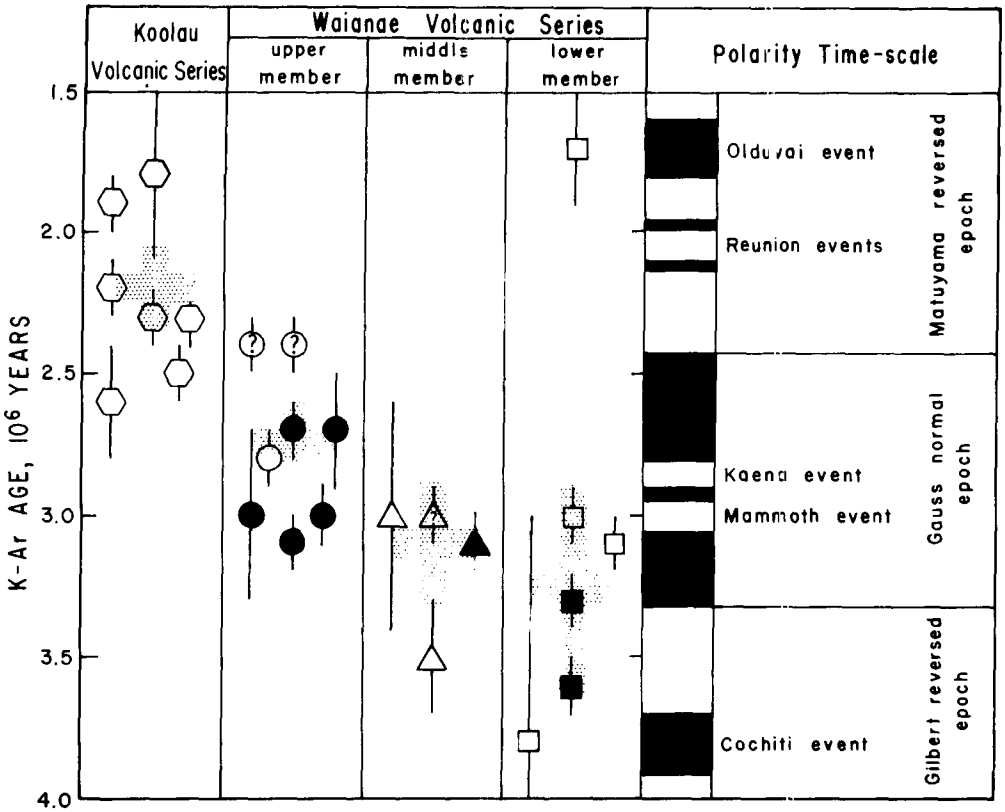


Figure 6. "Best" potassium-argon ages and magnetic polarities for the Koolau and Waianae Volcanic Series, Oahu, Hawaii (Table 4). Vertical bars are the estimated standard deviations of analytical precision. Shaded crosses indicate weighted mean and corresponding standard deviation (see text). Other symbols as in Figure 1. Polarity time scale after Cox (1969b) and Dalrymple (1972).

Matuyama reversed polarity epoch and is therefore Pliocene(?) and Pleistocene.

The ages in Figure 6 suggest that the subaerial part of Waianae Volcano was built within a period of about 1.0 to 1.2 m.y. or less. The lower and middle members may be approximately contemporaneous within the limits of the potassium-argon ages and were followed within a few hundred thousand years or less by the upper member, which forms the thin alkalic cap of the volcano. The subaerial part of Koolau Volcano was formed almost entirely after the Waianae Volcano, but there is some indication, in particular the ages of 2.4 m.y. from sites T and W, that Waianae Volcano was still active when Koolau volcano emerged from the sea. The subaerial part of Koolau Volcano was active from about 2.6 to about 1.8 m.y. ago. This interval of 0.8 m.y. is roughly the same as the interval of time required to construct the subaerial Waianae Volcano and also the same as the interval suggested by potassium-argon ages from lavas of Kohala Volcano on the island of Hawaii (McDougall, 1969; Dalrymple, 1971). Although these intervals are short and imply rapid rates of extrusion, they are consistent with other geological evidence. Jackson and Wright (1970), for example, have estimated the volume of the subaerial and submarine Koolau Volcano to be 45,000 km^3; if the historic eruption rate for all volcanoes on the island of Hawaii is typical, then the entire Koolau shield could have been built up from the sea floor in only 0.9 m.y. From eruption rates during the last two decades, Swanson (1972) has estimated that Kilauea Volcano on Hawaii could have been constructed within the last 0.4 m.y. Macdonald and Abbott (1970) have suggested that the entire submarine and subaerial mass of Mauna Loa Volcano could have been built in 1.5 m.y. if it erupted at its historic rate throughout its lifetime. Thus the

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

available evidence suggests that the tholeiitic shield-building phase of Hawaiian volcanoes is very short, lasting from about 0.5 to no more than 1.5 m.y.

Some information about the extrusion rates of Hawaiian shield volcanoes can be found in the data from the three thick sections sampled at sites N, O, and B. The fact that rocks of opposite polarity were never found at the same site suggests that the lavas of each section were extruded during a period of uniform polarity. If this is true, then the maximum length of time required to form the lavas at any of these sites is 0.4 m.y., the duration of the longest period of uniform polarity within the lifetime of Koolau and Waianae Volcanoes. The lower Waianae lavas sampled at site N represent a total stratigraphic thickness of 470 m and have normal polarity. Potassium-argon ages on two flows from the top and bottom of this site range from 3.2 to 4.0 m.y. and have a "best" age of about 3.6 ± 0.1 m.y. (Table 4). If we assume that these lavas were extruded during a single interval of normal polarity, then the most likely normal interval is the Cochiti event, which lasted only about 0.22 m.y. (Cox, 1969b). The only other possibility is the 0.26-m.y.-long normal period during the first part of the Gauss normal epoch. Assuming an average flow thickness of 2 m, extrusion rates of one flow every 900 to 1,100 yrs are indicated.

At site O, which is mostly in middle Waianae lavas, a thickness of about 360 m was sampled. This entire section has reversed polarity. Potassium-argon ages on middle Waianae lavas from this section range from 2.6 to 3.6 m.y. (Fig. 5) and have a "best" age of 3.0 ± 0.4 m.y. If one age of 3.6 m.y. (Funkhouser and others, 1968) is excepted, the available potassium-argon ages indicate that the lavas of this section were extruded between about 2.6 and 3.1 m.y. ago. Within this age range, there are only two brief reversed polarity events, the Kaena and the Mammoth, which are about 0.10 and 0.12 m.y. long, respectively (Cox, 1969b). If the section accumulated during both of these reversed events and if the brief (0.04 m.y.) intervening normal period is unrepresented or unsampled, then the maximum time for the formation of this section is 0.26 m.y. For flows averaging 2 m thick, eruption rates of one flow every 500 to 1,400 yrs are indicated.

The length of time required to form the 190-m-thick section of Koolau lavas at site B cannot be estimated so closely, but the available data suggest that it accumulated within an interval of about 0.4 m.y. or less, or about one flow of 2-m thickness every 4,200 yrs. These periods of 500 to 4,200 yrs, with the best estimates between 500 and 1,400 yrs, are in good agreement with the estimate of Doell and Cox (1965) that 1,000 yrs may be a realistic figure for the average time between superimposed flows of Mauna Loa and Kilauea and further indicate that Hawaiian shield volcanoes are built rapidly.

Reversal Time-Scale Data

One of the main reasons for this study, begun in 1965, was to obtain data for the geomagnetic reversal time scale, which was then in a stage of rapid development (Dalrymple, 1972). One of the earliest applications of joint potassium-argon and paleomagnetic research to the geomagnetic-reversal problem was by McDougall and Tarling (1963), who contributed 12 new paleomagnetic/potassium-argon data from the Hawaiian Islands, including Oahu. In order to avoid confusion caused by the incorporation of uncertain data, minimum standards for reversal data eventually were proposed and adopted (Doell and others, 1966; McDougall and Chamalaun, 1966; Cox and Dalrymple, 1967). One of these standards was that the samples used for dating and paleomagnetism must be collected from the same cooling unit. Although McDougall and Tarling could relate sample localities, they had worked mostly on different samples collected at different times, and unfortunately much of their 1963 data had to be rejected.

The data of McDougall and Tarling (1963) clearly indicated that Oahu was a location of abundant volcanic rocks that formed within a time range of interest for the geomagnetic-reversal problem. Unfortunately, the low precision of our potassium-argon data, discussed above, means that most of the results of our study do not fit the minimum requirements of dating precision (estimated standard deviation of 0.1 m.y. or 5 percent) for reversal time-scale data.[1] Seven ages are adequate, however, and these are listed by unit and site number in Table 5. Included in this list are new data on the massive hawaiite flow that led to the discovery, by McDougall and Chamalaun (1966), of the Kaena reversed event within the Gauss normal polarity epoch. Because of the im-

[1] Note that the data of Table 4 and Figure 6 are not suitable for defining the reversal time scale because they are "best" mean ages and polarities and therefore do not meet the criteria for acceptability.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

portance of this magnetic event, we collected and measured samples from this same flow (unit 62). Paleomagnetic measurements on eight oriented cores (Table 1) and potassium-argon ages on three of these cores (Table 3) confirm the earlier results. These new data, when averaged with the ages previously published by McDougall and Chamalaun (1966) and McDougall and Ur-Rahman (1972), give an age of 2.85 ± 0.05 m.y. Units 19 and 20 from site O in the middle member of the Waianae Volcanic Series also may have formed during the Kaena event, but data from this locality are slightly discrepant (Table 3), and these rocks may instead have formed during the Mammoth event. The four flows with normal polarity (units 52, 56, 59, and 61) in Table 5 fall within the Gauss normal epoch, as presently defined.

TABLE 5. NEW POTASSIUM-ARGON AND POLARITY DATA THAT MEET THE MINIMUM REQUIREMENTS FOR THE GEOMAGNETIC POLARITY TIME SCALE

Unit no.	(Site)	Potassium-argon age (10^6 yrs)	Standard deviation of precision (10^6 yrs)	Polarity
19	(O)	2.63	0.06	R
20	(O)	2.77	0.08	R
52	(Y)	2.79	0.07	N
56	(J)	3.30	0.10	N
59	(K)	3.10	0.10	N
61	(K)	3.06	0.08	N
62, GA1128*	(Z)	2.85	0.05	R

* Based on work by McDougall (1964), McDougall and Tarling (1963), McDougall and Ur-Rahman (1972), and the authors (this paper) on the same flow.

Paleomagnetic Secular Variation

At most localities over the surface of the Earth, about 80 percent of the strength of the Earth's magnetic field can be ascribed to an inclined geocentric magnetic dipole that best fits the observed field directions all over the Earth. This dipole is called the best-fit inclined dipole or main dipole. The remaining part of the field is called the nondipole field; it changes its shape and moves slowly in a westward direction, producing magnetic field changes at any locality with quasi-periods of a few hundreds to a few thousands of years. The main dipole field also changes its orientation and intensity, but with quasi-periods about an order of magnitude longer, or about 10^4 yrs, than those characteristic of nondipole changes. Together, these two phenomena produce geomagnetic secular variation, which has been discussed in detail by Doell and Cox (1971, 1972).

In order to calculate a reliable mean direction for the magnetic field from a set of paleomagnetic data or to estimate the paleosecular variation from a set of paleomagnetic directions, it is necessary that the lava flows from which the data are obtained span several of the longest periods involved in secular variation. For mean directions, this is necessary so that secular variation will be averaged out; for paleosecular variation measurements, this is necessary so that the full range of secular variation will be sampled. For formations like the Waianae and Koolau Volcanic Series, this requirement is generally satisfied, because it is unlikely that an entire series would be formed in less than a few tens of thousands of years.

The statistical methods used to determine the accuracy of a mean direction (Fisher, 1953) or of paleosecular variation (Cox, 1969a) require that the data be randomly drawn from the entire population of possible paleomagnetic directions for the period of time in question. For a stacked sequence of lavas, some of the superposed units may have been extruded in rapid succession and may have similar magnetic directions; this is called serial correlation. When this happens, the assumption of randomness is not satisfied, and the statistical parameters are, in the strict sense, invalid. This does not mean that the values for mean paleomagnetic direction or paleosecular variation are in error, but only that the formally calculated confidence intervals about these values are too small and give an erroneous impression of accuracy.

Although it is not possible to make a quantitative estimate of randomness, Watson and Beran (1967) have developed a test for serial correlation that gives a qualitative measure of randomness in a given set of ordered magnetic directions. Their test is to sum the cosines of the angles between successive directions as observed and then to compare this sum with similar sums computed from a large number of randomly generated sequences made from the same set of directions. The amount by which the sum of cosines in the actual sequence exceeds the mean value of the randomly generated sums is a qualitative measure of the

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

degree to which the observed data fail the assumption of randomness.

This test was performed on three sequences of Waianae Volcanic Series lavas (sites N, O, and X-Y) and one in the Koolau Volcanic Series (site B) (Table 6). Serial correlation is indicated at the 95-percent level of confidence for the sections at sites N, X-Y, and B, and so the formally calculated confidence intervals for these sites probably are slightly smaller than the true limits.

The paleosecular variation observed in the lavas of the Waianae and Koolau Volcanic Series is given (Table 7) as dispersion of RM directions about means, dispersion of VGP's about means, dispersion of RM directions about the axial geocentric dipole field direction in Hawaii, and dispersion of VGP's about the rotation axis. In order to obtain the paleosecular variation from the data of Tables 1 and 2, it is necessary to remove not only the dispersion caused by analytical error and natural within-flow variations but also the contribution of local magnetic anomalies. An "anomaly angular dispersion" figure of 1.25°, previously determined for the island of Hawaii (Doell and Cox, 1963), was assumed for the Oahu calculations because the two islands should be quite similar in this regard.

Although any of the parameters subscripted "F" in Table 7 may be used as measures of paleosecular variation, we prefer to use the angular standard deviation of VGPs measured with respect to the geographic axis (S_F). The reasons for this choice (Doell, 1970; Doell and Cox, 1971, 1972) are that the rotation axis seems to be the long-term symmetry axis for the geomagnetic field (Cox and Doell, 1960; Irving, 1964) and that the rotation axis figures prominently in all modern theories for the origin of the Earth's magnetic field. If Wilson's (1970) hypothesis suggesting a northward displacement of the main axial dipole is correct (see following section), it might be better to use dispersion parameters about a VGP calculated from the field direction in Oahu that would be produced by a northward displaced axial dipole which best fits all world-wide paleomagnetic data of comparable age. This would be a relatively small refinement and would reduce the apparent angular dispersion.

Using Cox's (1969a) methods, the formally calculated 95-percent limits of confidence about the Koolau S_F of 12.1° are 10.4° and 14.5°; about the Waianae S_F of 14.5°, they are 12.8° and 16.7°; and for the combined data S_F of 13.6°, the limits are 12.3° and 15.2°. These are minimum ranges because of the serial correlation observed in some of the data, but the true limits probably are not greatly different.

On the basis of observatory measurements, Fisk (1931) found that the rate of change in the Earth's magnetic field in the central Pacific is anomalously low. From paleomagnetic studies on historic and prehistoric lava flows from the island of Hawaii, Doell and Cox (1963, 1965, 1971) and Doell (1972a) have shown that this low secular variation has persisted in the central Pacific for at least 700,000 yrs. They attribute this phenomenon to a virtual absence of the nondipole field in this region. The cause is not known, but it may be related to some inhomogeneity in the lower mantle or to irregularities in the core-mantle boundary (Cox, 1962; Doell and Cox, 1971). The length of time that this condition has persisted in the central Pacific is clearly of interest. Tarling (1965) has suggested that the relative lack of a nondipole component in the Pacific area may be a recent phenomenon. In contrast, Doell (1972b, 1972c) has concluded that the relative low has persisted for at least 5 to 10 m.y.

We have plotted the available Hawaiian angular dispersion data (S_F) as a function of time in Figure 7. Values from France, Iceland, the western United States, Galapagos, Alaska, and Antarctica are shown for comparison. It is clear that Hawaii has been an area of anomalously low secular variation during the Brunhes normal polarity epoch (0.7 m.y. to the present)

TABLE 6. SERIAL CORRELATION ANALYSIS

Site	Units	No. of lavas	Sum of cosines of observed sequence	Sum of cosines of random sequences mean	Sum of cosines of random sequences std. dev.	Difference between observed and random means in no. of standard deviation
N	01–13	11	21.6	21.3	0.1	3.0
O	14–42	18	7.86	7.79	0.06	1.2
X and Y	43–51	9	16.4	15.9	0.2	2.5
B	76–98	23	9.68	9.40	0.14	2.0

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 7. DISPERSION STATISTICS OF FLOW-AVERAGE DATA

Data analyzed	Analyses of VGP's								
	N	S_T	S_W	S_B	S_F	K_T	K_W	K_B	K_F
Referred to means									
Koolau Volcanic Series	7.82	11.4°	6.8°	11.2°	11.0°	50	117	54	55
Waianae Volcanic Series	7.78	13.6	7.8	13.4	13.2	36	78	39	40
All data combined	7.79	12.8	7.4	12.5	12.4	40	89	44	45
Referred to axial geocentric dipole									
Koolau Volcanic Series	7.82	12.4°	6.8°	12.2°	12.1°	44	117	46	47
Waianae Volcanic Series	7.78	14.9	7.8	14.6	14.5	31	78	32	33
All data combined	7.79	14.0	7.4	13.7	13.6	34	89	36	37

N, mean number of sample cores per lava flow.
S, s, angular standard deviation for VGP's and RM directions, respectively.
K, k, precision parameters for VGP's and RM directions, respectively.
T, total dispersion.

when compared with the expected value of 14°, which was calculated from the known nondipole component of the 1965 geomagnetic field and a component due to a dipole wobble of 11° (Doell and Cox, 1971). In contrast, data of comparable age from the other localities fall within the expected values for their respective latitudes on the basis of the present nondipole field and the 11° wobble.

It can be seen from Figure 7, however, that the angular dispersions for all localities increase with age. There are at least three hypotheses to explain this increase with age: (1) that the increase is only apparent and is caused by the increasing effects of weathering, tectonism, and geomorphic processes as rocks grow older; (2) that the long-term mean ratio of dipole to nondipole field strengths has been increasing, which would lead to a decreasing observed angular dispersion; and (3) that secular variation during the last several million years has been slowly decreasing because of decreasing wobble of a more or less constant mean strength dipole with the present mean ratio of dipole strength to nondipole strength. If any of these hypotheses explains the angular dispersion decrease during the last 5 m.y., then it is suggested that the expected values indicated in Figure 7 should increase with age and the proposed persistence of the central Pacific nondipole field low, relative to other localities, is still valid.

VGP Positions

The mean VGPs and RM directions for the Koolau Volcanic Series, the Waianae Volcanic Series, and both series combined are given in Table 8 with appropriate statistical parameters. Not all data in Tables 1 and 2 were used in calculating these means. VGPs more than 44° from the rotation axis have not been used in the calculations, because these probably represent a different type of phenomenon (VGP excursions, which are discussed later) and would have an inordinate influence on the apparent VGP position and angular dispersion (Doell and Cox, 1971). To further maintain statistical uniformity, all flow-average data with α_{95} greater than 9° have been excluded.

The 95-percent limits of confidence about the paleomagnetic poles for the Koolau and Waianae Volcanic Series are shown in Figure 8. Both paleomagnetic poles are just significantly different from the rotation axis, but because of the serial correlation found for some of the sites, it is unlikely that this difference is significant in the statistical sense. It is interesting to note, however, that the two paleomagnetic poles are on the side of the rotation axis away from Oahu. Wilson (1970) has noted this "opposite side" feature in many paleomagnetic studies of Tertiary and Quaternary rocks. He has suggested that it might indicate a permanent displacement of the average geomagnetic axial dipole northward from the Earth's center along the rotation axis. Although probably not significant at the 95-percent confidence level, the mean VGPs from Oahu do lend support to this hypothesis, as do other paleomagnetic data from the Hawaiian Islands (Doell and Cox, 1965; Doell, 1969, 1972a, 1972b, 1972c). This "opposite side" effect apparently was a relatively permanent feature of the paleomagnetic pole in the central Pacific during the past several million years.

VGP Excursions

Occasionally, the Earth's magnetic field appears to undergo large, rapid excursions in direction. These excursions do not appear to be

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

TABLE 7. *Continued*

Data analyzed	Analyses of RM directions							
	s_T	s_W	s_B	s_F	k_T	k_W	k_B	k_F
Referred to means								
Koolau Volcanic Series	14.8°	4.7°	14.7°	14.6°	30	269	31	31
Waianae Volcanic Series	17.9	5.6	17.8	17.7	21	180	22	22
All data combined	16.7	5.3	16.6	16.6	24	206	25	25
Referred to axial geocentric dipole								
Koolau Volcanic Series	17.0°	4.7°	16.9°	16.8°	24	269	24	24
Waianae Volcanic Series	20.6	5.6	20.5	20.5	16	180	16	17
All data combined	19.2	5.3	19.2	19.1	18	206	19	19

W, within flow dispersion.
B, between flow dispersion.
F, ancient geomagnetic field dispersion.

part of ordinary secular variation but instead are abrupt, "spike-like" phenomena of much larger amplitude. Three such excursions have been recorded in superposed lavas from Oahu—two from the Waianae Volcanic Series (site N, units 1–4; site O, units 30–36) and one from the Koolau Volcanic Series (site F, units 69–72). At site N, the VGPs seem to be approaching the present rotation axis from low latitudes (Fig. 9). At site O, the VGP's move from a near polar position to just south of the equator and back again to high latitudes. The four Koolau Volcanic Series lavas sampled at site F suggest that the VGP was in the process of leaving the polar region for low latitudes.

That these excursions of the VGP are rare events may be inferred from that fact that only 5 out of 99 units sampled have VGPs more than 44° from the rotation axis. They are even less common in other paleomagnetic studies of the Hawaiian Islands.

There are several hypotheses to explain these VGP excursions: (1) they are due to rapid tilts of the main dipole, (2) they represent aborted reversals of geomagnetic polarity, or (3) they are the result of nondipole field changes. It is not possible to choose between hypotheses 1 and 2 without paleointensity data, but a nondipole-field origin seems unlikely for two reasons. First, the excursions appear to be along great-circle paths (Fig. 9), which suggests a geocentric phenomenon. Second, other paleomagnetic data from the Hawaiian Islands (Doell, 1972a; Doell and Cox, 1965) indicate that the nondipole field in this region is largely absent. Whatever their cause, these excursions, although rare, are interesting features of the geomagnetic field.

APPENDIX 1. TECHNIQUES

A total of 786 cores was collected from 99 units (that is, lava flows or intrusive bodies) at 18 sites—11 in the Waianae Range and 7 in the Koolau Range. (In this paper, the term "site" is used to indicate a sampling location or exposure that may consist of more than a single lava flow. In some

TABLE 8. STATISTICAL ANALYSES OF FLOW-AVERAGE DATA

Data analyzed	Lat	Long.	N	R	α_{95}	K_T	S_T
Koolau Volcanic Series VGP's	85.2° N.	38.7° E.	33	32.365	3.6°	50	11.4°
Waianae Volcanic Series VGP's	84.1° N.	34.8° E.	55	53.485	3.3°	36	13.6°
ALL VGPs	84.5° N.	36.1° E.	88	85.845	2.4°	40	12.8°
	Incl.	Decl.	N	R	α_{95}	k_T	s_T
Koolau Volcanic Series RM directions	29.8°	358.9°	33	31.943	4.6°	30	14.8°
Waianae Volcanic Series RM directions	28.1°	358.7°	55	52.430	4.3°	21	17.9°
All RM directions	28.8°	358.8°	88	84.363	3.2°	24	16.7°

N, number of units.
R, vector sum of N unit vectors.
α_{95}, 95 percent confidence semiangle about means.
K_T, k_T, total precision parameter, VGPs and RM directions, respectively.
S_T, s_T, total angular standard deviation about means, VGPs and RM directions, respectively.
All reversed flow average data converted to normal, north-seeking data before analysis.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

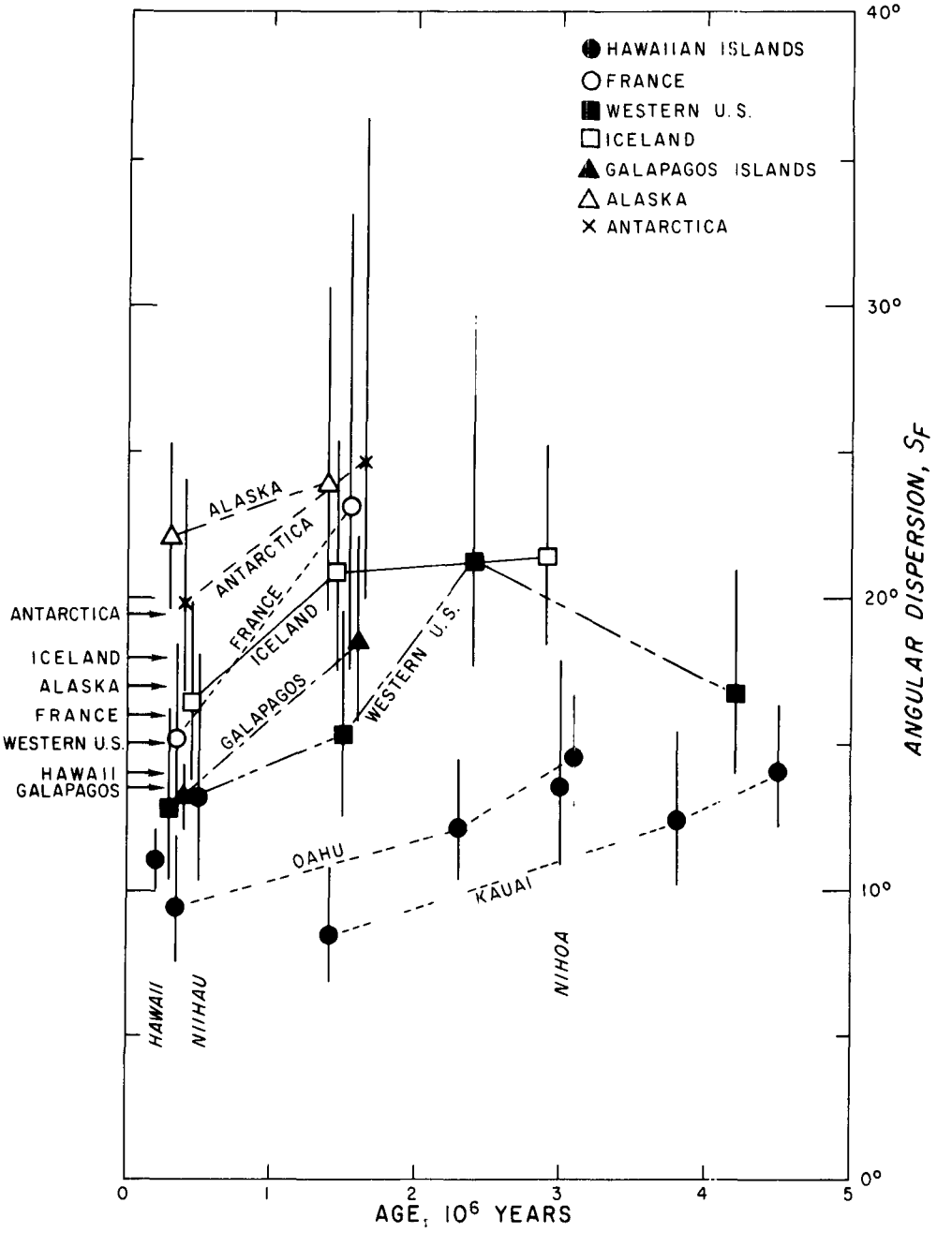


Figure 7. Change in secular variation (angular dispersion, S_F) with time for the Hawaiian Islands. Included for comparison are values for Iceland, France, western United States, Galapagos, Alaska, and Antarctica. Ages are averages of radiometric data or are plotted at the approximate midpoints of geomagnetic polarity epochs. Arrows show the expected values of present angular dispersion calculated for the latitudes of the seven localities on the basis of an 11° dipole wobble and the nondipole component of the 1965 IGRF (Doell and Cox, 1971). Lines connect data from the same island or area. Vertical bars are 95-percent limits of confidence (Cox, 1969a). Data from Doell (1970, 1972a, 1972b, 1972c, 1972d), Doell and Cox (1965), Cox (1971 and unpub. data), and this paper.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

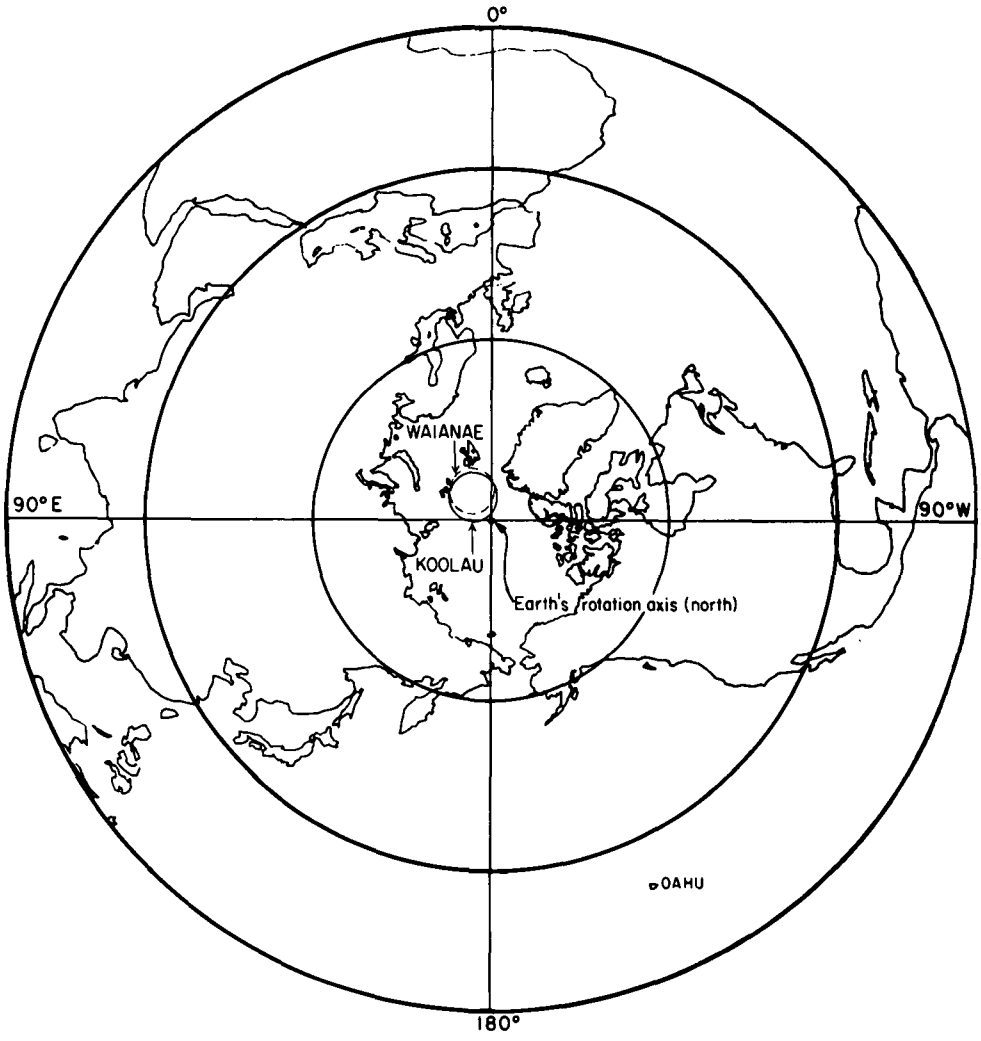


Figure 8. Ninety-five-percent limits of confidence for the mean VGPs calculated from paleomagnetic measurements of 55 lavas of the Waianae Volcanic Series (dashed circle) and 33 lavas of the Koolau Volcanic Series (full circle). Lambert equal-area projection.

literature on paleomagnetism, site refers to a single lava or other cooling unit. Here the term "unit" is used in this sense.) The sample sites, designated by letters, are shown in Figure 1 and described in Appendix 2; they were chosen on the basis of accessibility, degree of exposure and weathering of the volcanic rocks, and the availability of complementary petrologic and age data of other workers. Units are designated by cardinal numbers from 1 to 99. At a given site, the unit numbers are arranged in stratigraphic order, the youngest unit having the largest number. Between sites, however, the stratigraphic relations are often unknown, and the unit numbers have no stratigraphic significance. Except for units 63 and 65, which are small intrusive bodies, all units sampled are lava flows.

At some sites, it was impractical to sample every unit, especially at Nanakuli Valley in the Waianae Range and Makapuu Head in the Koolau Range where hundreds of flows are exposed in section. At these places, short, preliminary, partially oriented cores were taken from most of the exposed flows. A thin section was made in the field with portable equipment (Dalrymple and Doell, 1966), and the preliminary cores were analyzed for polarity with a field fluxgate magnetometer (Doell and Cox, 1962). These preliminary polarity and petrographic data were then used to select the units from which oriented cores for paleomagnetism and dating were

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

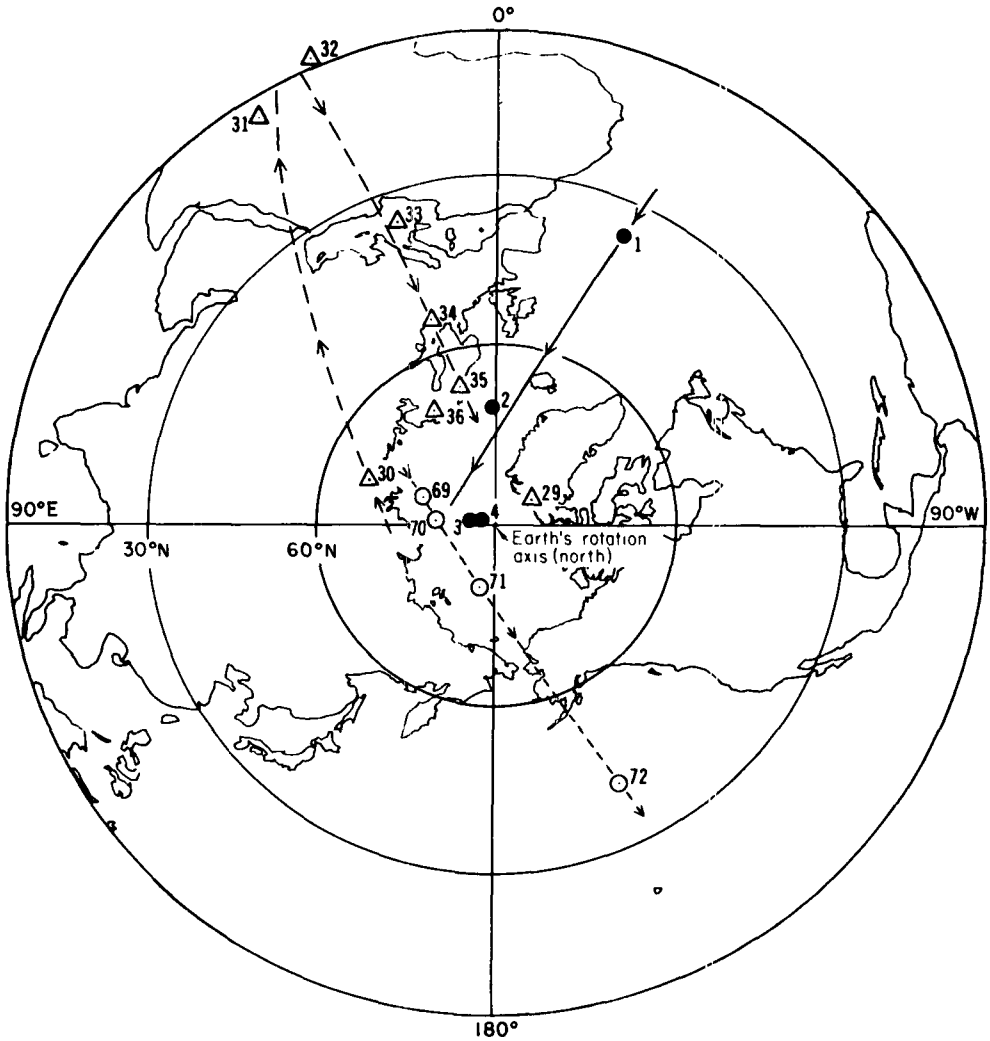


Figure 9. VGP excursions recorded in paleomagnetic measurements from three sets of superposed lava flows. Filled circles are from site N, and open triangles are from site O in the Waianae Volcanic Series; open circles are from site F in the Koolau Volcanic Series. Open symbols are north-seeking VGPs, closed symbols are south-seeking VGPs, and numbers are unit numbers. Lambert equal-area projection.

collected. This procedure increased significantly the percentage of collected units that were suitable for potassium-argon dating.

Cores were drilled and geographically oriented in the field using portable equipment (Doell and Cox, 1967). In order to get an average value of the remanent magnetic direction for the entire flow (Doell and Cox, 1963), seven to nine cores (usually eight) of 2.5-cm diameter were collected from each unit over a lateral span of 10 m or more where possible. One 2.25-cm-long specimen from each core was used for magnetic measurement. Doell and Cox (1963) have shown that paleomagnetic measuring techniques are significantly more precise than both the orienting procedure and the natural differences in magnetic directions between cores from a typical lava flow, and so there is no significant gain in accuracy if more than one specimen from each core is measured. The direction and intensity of the remanent magnetism (RM) in each specimen was measured using spinner magnetometers. After the natural remanent magnetization (NRM) was measured, one or two specimens from each unit was progressively demagnetized in alternating fields (AF) of 25, 50, 100, 200, and 400 (and sometimes 600 and 800) oersteds peak field strength. Units for which the NRM directions were tightly grouped and did not change during progressive AF de-

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

magnetization were not magnetically cleaned further. For these units, the mean NRM directions are obviously not influenced by anomalous RM and are a reasonably precise estimate of the original thermoremanent magnetization (TRM) direction. For the units where NRM directions were initially scattered, the results of the progressive AF demagnetization were used to choose peak AF strengths that would remove the anomalous magnetization that causes the scatter. The remaining specimens from such a flow were then demagnetized at the appropriate AF strength in order to obtain a tight grouping of directions. Additional discussion of the laboratory and field equipment and techniques are given by Doell and Cox in a series of articles in Collinson and others (1967). Statistical techniques for the treatment of paleomagnetic data are described by Fisher (1953), Watson and Irving (1957), Cox (1970), and Doell (1970).

Laboratory-prepared thin sections of one core or more from each unit were examined in order to determine which units were suitable for potassium-argon dating, according to the criteria explained by Dalrymple and Lanphere (1969). The potassium content of most of the suitable samples was measured so that flows with unusually low potassium (and hence low argon) could be avoided where possible. The final selection of samples for dating was made on the basis of petrographic suitability, potassium content, and stratigraphic position. One to three cores (designated a, b, and c) from each unit were dated. These were cut into blocks weighing about 10 to 12 grams each. A small slice was taken from each end of the block, the slices were pulverized, and the powder was mixed together and then split into two aliquots with a Jones-type microsplitter. The potassium content of each powdered split was measured using the lithium metaborate fusion technique and flame photometry (Suhr and Ingamells, 1966; Ingamells, 1970). One or two argon analyses were made on the remaining larger blocks from the core using isotope dilution, a bulb-type Ar^{38} tracer system, and both Reynolds-type 4½-in. radius and Nier-type 6-in. radius rare-gas mass spectrometers operated in the static mode. The argon analysis and data reduction procedures are discussed fully in Dalrymple and Lanphere (1969).

APPENDIX 2. LOCATIONS AND DESCRIPTIONS OF SITES IN TABLES 1 AND 2 AND JUSTIFICATION OF "BEST" POTASSIUM-ARGON AGES AND MAGNETIC POLARITIES OF TABLE 4

Site A (units 73–74): Two superposed lava flows on northeast side of Waimea Bay, lat 21°38′43″ N., long. 158°03′55″ W., sampled in roadcuts 75 m west of point "251," Waimea 7½′ quadrangle (1952). Our two measurements do not agree within analytical precision, but the older age of 2.32 m.y. agrees with an age of 2.30 m.y. reported by McDougall (1964) and McDougall and Ur-Rahman (1972). Accordingly, an age of 2.3 ± 0.1 has been used for this site. The lavas are reversely magnetized.

Site B (units 76–98): Twenty-three superposed lavas sampled in the cliff 100 m southwest of Makapuu Beach, Koko Head 7½′ quadrangle (1959). Starting (unit 76) in gully 300 m west of BM 144 at 100-ft elevation, lat 21°18′46″ N., long. 157°39′45″ W., and ending (unit 98) at 730 ft elevation atop ridge, lat 21°18′43″ N., long. 157°39′58″ W. Potassium-argon measurements on five samples from four flows (Table 3) show a mean age of 1.94 m.y. Only samples 76a and 98 repeat satisfactorily, and we consider their average age, 1.78 ± 0.26 m.y., the best estimate for this locality. All 23 lavas have reversed polarity.

Site C (units 67–68): Two superposed lava flows in roadcut E side of Dole Ave., 3,550 ft east of intersection with University Ave., lat 21°17′52″ N., long 157°48′50″ W., Honolulu 7½′ quadrangle (1959). Duplicate argon measurements on unit 67 agree within 1.5 percent. The calculated age of 2.56 ± 0.18 m.y. has been averaged with one of 2.43 m.y. obtained by McDougall (1964) and McDougall and Ur-Rahman (1972) to give an age of 2.5 ± 0.1 m.y. for this locality. The two flows have reversed polarity.

Site D (unit 66): Lava flow sampled in roadcut along Nuuanu Pali Highway, 380 m northeast of BM 96, 200-ft elevation, lat 21°19′31″ N., long. 157°50′58″ W., Honolulu 7½′ quadrangle (1959). Potassium-argon data for two samples from this flow are inconsistent, but the age of 2.60 ± 0.17 m.y. has slightly better repeatability. The lava is reversely magnetized.

Site E (unit 99): Lava flow in roadcut southwest side of Moanalua Road, 300 m northeast of Salt Lake, 80-ft elevation, lat 21°21′33″ N., long. 157°54′5″ W., Puuloa 7½′ quadrangle (1959). No age data are available for this reversely magnetized unit.

Site F (units 69 through 72): Four superposed lavas sampled in the Pacific Coast Aggregates Quarry immediately east of North Halawa Stream about 3.2 km east of Aiea, lat 21°22′47″ N., long. 157°54′32″ W., Waipahu 7½′ quadrangle (1959). The age of 1.86 m.y. for both units 69 and 72 do not agree with the one of 2.32 m.y. found by McDougall (1964) and McDougall and Ur-Rahman (1972). Because the later age is based on a single lava, the age of this locality is taken as 1.86 ± 0.11 m.y. It may be, however, that McDougall's data were obtained from a different lava flow. All four units are reversely magnetized.

Site G (unit 75): Lava in roadcut on east side of Moanalua Road about 1.6 km northwest of McGrew Point, 300 m north of Kamehameha Highway, lat 21°23′25″ N., long. 157°57′00″ W., Waipahu 7½′ quadrangle (1959). Our age data on this unit are inconsistent compared to those of

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

McDougall (1964) at this locality. His age of 2.17 ± 0.07 m.y. is preferred. Unit 75 is reversely magnetized.

Site H: McDougall (1964) obtained a value of 2.31 ± 0.07 m.y. from a single age measurement at this locality, and McDougall and Tarling (1963) report reversed polarity.

Site I: Funkhouser and others (1968) obtained ages of 4.0 ± 0.8 and 3.5 ± 0.8 m.y. on a sample from the 1,081-ft level of a drill hole at this locality. Their mean age of 3.8 ± 0.8 m.y. is used. The magnetic polarity is not known.

Site J (unit 56): Lava flow sampled in roadcuts on north side of Highway 90 about 50 m west of junction of Kalaeloa Road, lat 21°20′14″ N., long. 158°05′33″ W., Ewa 7½′ quadrangle (1962). Single ages by us and Funkhouser and others (1968) differ significantly. Since the two samples are probably in proximity, the best estimate of the age is the mean of 3.10 ± 0.1 m.y. Unit 56 is normally magnetized.

Site K (units 59 through 61): Three superposed lava flows in roadcuts on north side of Highway 90 about 2.4 km east-southeast of Kahe Point, lat 21°20′44″ N., long. 158°06′45″ W., Ewa 7½′ quadrangle (1962). Three lavas, units 59 and 61 and one from McDougall (1964), have been dated. The stratigraphic relation of McDougall's sample to units 59 through 61 is not known, but all three ages were based on multiple measurements, and all possess normal polarity (Table 1; McDougall and Tarling, 1963). The mean of 3.00 ± 0.04 m.y. is the best estimate of the age of lavas at this site.

Site L (units 57 and 58): Two superposed lavas in roadcuts on north side of Highway 90, about 1.3 km southeast of Kahe Point, lat 21°20′52″ N., long. 158°07′23″ W., Ewa 7½′ quadrangle (1962). A normally magnetized lava at this locality (McDougall and Tarling, 1963) was dated by McDougall (1964) as 3.28 ± 0.12 m.y. old. Units 57 and 58 are also normally magnetized.

Site M: An age of 2.74 ± 0.04 m.y. was obtained by McDougall (1964) on a single flow. Its relation to the normally magnetized lava reported by McDougall and Tarling (1963) is not known.

Site N (units 1 through 13): Thirteen superposed lavas sampled from outcrops on nose in southeast side of Nanakuli Valley, starting (unit 1) at lat 21°23′48″ N., long. 158°07′12″ W., 500-ft elevation, and ending (unit 13) at lat 21°23′50″ N., long. 158°06′59″ W., 1,500-ft elevation, Schofield Barracks 7½′ quadrangle (1960). This is the same section measured and sampled by Macdonald and Katsura (1964). Age measurements were made on two samples from each of units 7 and 13; the average age is 3.62 ± 0.11 m.y. All 13 lavas have normal polarity.

Site O (units 14 through 42): Twenty-nine superposed lavas sampled on nose on north side of Nanakuli Valley, starting (unit 14) at lat 21°24′26″ N., long. 158°07′20″ W., 820-ft elevation, and ending (unit 42) at lat 21°24′58″ N., long. 158°07′17″ W., 2,100-ft elevation, Schofield Barracks 7½′ quadrangle (1960). There is a breccia between units 18 and 19, separating the middle and lower members of the Waianae Volcanic Series (Stearns and Vaksvik, 1935). The single age of 1.70 ± 0.15 m.y. obtained on unit 15 from the lower member is clearly anomalous and should be considered uncertain. Funkhouser and others (1968) dated two lavas from the middle member, and we have dated units 19, 20, 31, and 41. The mean age for the six flows (excluding the data for sample 41a, which did not have repeatable potassium-argon measurements) is 3.00 ± 0.38 m.y. Except for the intermediate directions, all lavas in this section have reversed polarity.

Site P (unit 64): Lava flow sampled in roadcuts off Kolekole Pass Road about 660 m south-southeast of Puu Kailio, lat 21°28′02″ N., long. 158°07′17″ W., 1,000-ft elevation, Schofield Barracks 7½′ quadrangle (1960). McDougall (1964) obtained an age of 3.46 ± 0.17 m.y. at this locality, but the relation to our reversely magnetized unit is not known.

Site Q (unit 63): Samples from southwest side of Mauna Kuwale Dome, lat 21°27′46″ N., long. 158°09′36″ W., 380-ft elevation, Waianae 7½′ quadrangle (1954). Extensive studies by Funkhouser and others (1968) have shown that the postassium-argon age of this rhyodacite is 2.3 ± 0.4 m.y. and that the 8.4-m.y. age determined by McDougall (1964) on a biotite was probably due to excess argon. The unit has normal polarity.

Site R: Multiple measurements by McDougall (1964) on a sample from this locality give an age of 3.02 ± 0.04 m.y. The magnetic polarity is unknown.

Site S: Ages from this site are internally inconsistent and should be considered unreliable (Funkhouser and others, 1968). The magnetic polarity is unknown.

Site T: An average age of 2.4 ± 0.1 m.y. has been determined on feldspar and whole-rock samples from this site by Funkhouser and others (1968). Magnetic polarity is unknown.

Site U: An age of 2.95 ± 0.04 m.y. for this site has been determined by McDougall (1964). A nearby flow has reversed polarity.

Site V (units 54 and 55): Two lavas sampled between the beach and Highway 90 just south of Kaluakaniloa Stream, lat 21°32′53″ N., long. 158°14′35″ W., Kaena 7½′ quadrangle (1954). One sample from this locality has been dated by Funkhouser and others (1968) and three samples (one from unit 55 and two from unit 54) have been dated by us. The mean age of the two samples with duplicate ages, 3.06 ± 0.14 m.y., is considered the best age estimate. The polarity of units 54 and 55 is reversed.

Site W: An age of 2.4 ± 0.1 m.y. has been obtained by Funkhouser and others (1968) on

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

feldspar and partially separated whole rock. Magnetic polarity is unknown.

Site X (units 43 through 47, and 65): Five superposed lavas and a cross-cutting dike (unit 65) exposed in roadcuts along road leading to "relay tower" on ridge east-southeast of Kaena Point, about 3 km east-southeast of Kaena Point near 1,000-ft elevation, lat 21°33′40″ N., long. 158°14′18″ W., Kaena 7½′ quadrangle (1954). Duplicate measurements on units 43 and 46 give an average age of 3.14 ± 0.12 m.y. The flows and dike all have normal polarity.

Site Y (units 48 through 53): Six superposed lava flows sampled in exposures on ridge immediately west-northwest from "relay tower" about 3.2 km east-southeast of Kaena Point, lat 21°33′52″ N., long. 158°15′05″ W., 1,180-ft elevation, Kaena 7½′ quadrangle (1954). These flows are stratigraphically higher than those at site X. Ages from units 52 and 48 and from a flow sampled by McDougall (1964) at this locality are reasonably consistent. The mean age is 3.00 ± 0.32 m.y., and those sampled by us all have normal polarity.

Site Z (unit 62): Samples from large massive aa flow in quarry near Dillingham AFB, north side of Waianae Range, lat 21°34′46″ N., long. 158°12′56″ W., 70-ft elevation, Kaena 7½′ quadrangle (1954). One sample dated by McDougall (1964) and McDougall and Ur-Rahman (1972), and three by us from this locality are in good agreement and average 2.85 ± 0.05 m.y. Polarity is normal.

REFERENCES CITED

Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., 1967, Developments in solid earth geophysics 3, *in* Methods in paleomagnetism: Amsterdam, Elsevier, 609 p.

Cox, A., 1961, Anomalous remanent magnetization of basalt: U.S. Geol. Survey Bull. 1083-E, p. 131–160.

—— 1962, Analysis of present geomagnetic field for comparison with paleomagnetic results: Jour. Geomagnetism and Geoelectricity, v. 13, p. 101–112.

—— 1969a, Confidence limits for the precision parameter κ: Royal Astron. Soc. Geophys. Jour., v. 17, p. 545–549.

—— 1969b, Geomagnetic reversals: Science, v. 163, p. 237–245.

—— 1970, Latitude dependence of the angular dispersion of the geomagnetic field: Royal Astron. Soc. Geophys. Jour., v. 20, p. 253–269.

—— 1971, Paleomagnetism of San Cristobal Island, Galapagos: Earth and Planetary Sci. Letters, v. 11, p. 152–160.

Cox, A., and Dalrymple, G. B., 1967, Statistical analysis of geomagnetic reversal data and the precision of potassium-argon dating: Jour. Geophys. Research, v. 72, p. 2603–2614.

Cox, A., and Doell, R. R., 1960, Review of paleomagnetism: Geol. Soc. America Bull., v. 71, p. 645–768.

—— 1964, Long period variations of the geomagnetic field: Seismol. Soc. America Bull., v. 54, pt. B, p. 2243–2270.

Crow, E. L., Davis, F. A., and Maxfield, M. W., 1960, Statistics manual: New York, Dover Pubs., 288 p.

Dalrymple, G. B., 1968, Potassium-argon ages of recent rhyolites of the Mono and Inyo Craters, California: Earth and Planetary Sci. Letters, v. 3, p. 289–298.

—— 1971, Potassium-argon ages from the Pololu Volcanic Series, Kohala Volcano, Hawaii: Geol. Soc. America Bull., v. 82, p. 1997–2000.

—— 1972, Potassium-argon dating of geomagnetic reversals and North American glaciations, *in* Bishop, W. W., and Miller, J. A., eds., Calibration of hominoid evolution: Recent advances in isotopic and other dating methods as applicable to the origin of man: Edinburgh, Scottish Academic Press, p. 107–134.

Dalrymple, G. B., and Doell, R. R., 1966, Portable core drill and thin-section laboratory for field use: U.S. Geol. Survey Prof. Paper 550-B, p. B182–B185.

Dalrymple, G. B., and Lanphere, M. A., 1969, Potassium-argon dating: San Francisco, W. H. Freeman and Co., 258 p.

Doell, R. R., 1969, Paleomagnetism of the Kau Volcanic Series, Hawaii: Jour. Geophys. Research, v. 74, p. 4857–4868.

—— 1970, Paleomagnetic secular variation study of lavas from the Massif Central, France: Earth and Planetary Sci. Letters, v. 8, p. 352–362.

—— 1972a, Paleosecular variation of the Honolulu Volcanic Series, Oahu, Hawaii: Jour. Geophys. Research, v. 77, p. 2129–2138.

—— 1972b, Paleomagnetism of lava flows from Kauai, Hawaii: Jour. Geophys. Research, v. 77, p. 862–876.

—— 1972c, Paleomagnetism of volcanic rocks from Niihau, Nihoa, and Necker Islands, Hawaii: Jour. Geophys. Research, v. 77, p. 3725–3730.

—— 1972d, Paleomagnetic studies of Icelandic lava flows: Royal Astron. Soc. Geophys. Jour., v. 26, p. 459–479.

Doell, R. R., and Cox, A., 1961, Paleomagnetism of Hawaiian lava flows: Nature, v. 192, p. 645–646.

—— 1962, Determination of the magnetic polarity of rock samples in the field: U.S. Geol. Survey Prof. Paper 450-D, p. D105–D108.

—— 1963, The accuracy of the paleomagnetic method as evaluated from historic Hawaiian lava flows: Jour. Geophys. Research, v. 68, p. 1997–2009.

—— 1965, Paleomagnetism of Hawaiian lava flows: Jour. Geophys. Research, v. 70, p. 3377–3405.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Doell, R. R., and Cox, A., 1967, Paleomagnetic sampling with a portable coring drill, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 21–25.

—— 1971, The Pacific geomagnetic secular variation: Science, v. 71, p. 248–254.

—— 1972, The Pacific geomagnetic secular variation anomaly and the question of lateral uniformity in the lower mantle, *in* Robertson, E. C., Hays, J. F., and Knopoff, L., eds., The nature of the solid earth: New York, McGraw-Hill, Inc., p. 245–284.

Doell, R. R., Dalrymple, G. B., and Cox, Allan, 1966, Geomagnetic polarity epochs: Sierra Nevada data 3: Jour. Geophys. Research, v. 71, p. 531–541.

Fisher, R. A., 1953, Dispersion on a sphere: Royal Soc. London Proc., ser. A, v. 217, p. 295–305.

Fisk, H. W., 1931, Isopors and isoporic movement: Internat. Geodetic Geophys. Union Bull., no. 8, Stockholm Assembly, 1930, p. 280–292.

Funkhouser, J. G., 1966, The determination of a series of ages of a Hawaiian volcano by the potassium-argon method [Ph.D. thesis]: Honolulu, Univ. Hawaii, 168 p.

Funkhouser, J. G., Barnes, I. L., and Naughton, J. J., 1966, Problems in the dating of volcanic rocks by the potassium-argon method: Bull. Volcanol., v. 29, p. 709–717.

—— 1968, The determination of a series of ages of Hawaiian volcanoes by the potassium-argon method: Pacific Sci., v. 22, p. 369–372.

Graham, K.W.T., 1961, The re-magnetization of a surface outcrop by lightning currents: Royal Astron. Soc. Geophys. Jour., v. 6, p. 85–102.

Gramlich, J. W., Lewis, V. A., and Naughton, J. J., 1971, Potassium-argon dating of Holocene basalts of the Honolulu Volcanic Series: Geol. Soc. America Bull., v. 82, p. 1399–1404.

Holmes, A., 1965, Principles of physical geology (2d ed.): New York, Ronald Press, 1288 p.

Ingamells, C. O., 1970, Lithium metaborate flux in silicate analysis: Anal. Chim. Acta, v. 52, p. 323–334.

Irving, E., 1964, Paleomagnetism and its application to geological and geophysical problems: New York, John Wiley & Sons, Inc., 399 p.

Jackson, E. D., and Wright, T. L., 1970, Xenoliths in the Honolulu Volcanic Series, Hawaii: Jour. Petrology, v. 11, p. 405–430.

Jackson, E. D., Silver, E. A., and Dalrymple, G. B., 1972, The Hawaiian-Emperor chain and its relation to Cenozoic circumpacific tectonics: Geol. Soc. America Bull., v. 83, p. 601–618.

Macdonald, G. A., and Abbott, A. T., 1970, Volcanoes in the sea: Honolulu, Univ. Hawaii Press, 441 p.

Macdonald, G. A., and Katsura, T., 1964, Chemical composition of Hawaiian lavas: Jour. Petrology, v. 5, p. 82–133.

McDougall, I., 1963, Potassium-argon ages from western Oahu, Hawaii: Nature, v. 197, p. 344–345.

—— 1964, Potassium-argon ages from lavas of the Hawaiian Islands: Geol. Soc. America Bull., v. 75, p. 107–128.

—— 1969, Potassium-argon ages on lavas of Kohala Volcano, Hawaii: Geol. Soc. America Bull., v. 80, p. 2597–2600.

—— 1971, Volcanic island chains and sea floor spreading: Nature, v. 231, p. 141–144.

McDougall, I., and Chamalaun, F. H., 1966, Geomagnetic polarity scale of time: Nature, v. 212, p. 1415–1418.

McDougall, I., and Tarling, D. H., 1963, Dating of polarity zones in the Hawaiian Islands: Nature, v. 200, p. 54–56.

McDougall, I., and Ur-Rahman, A., 1972, Age of the Gauss-Matuyama boundary and of the Kaena and Mammoth events: Earth and Planetary Sci. Letters, v. 14, p. 367–380.

Moore, J. G., 1970, Relationship between subsidence and volcanic load, Hawaii: Bull. Volcanol., v. 34, p. 562–576.

Morgan, W. J., 1971, Convection plumes in the lower mantle: Nature, v. 230, p. 42–43.

—— 1972, Plate motions and deep mantle convection, *in* Shagam, R., ed., Studies in Earth and space sciences (Hess volume): Geol. Soc. America Mem. (in press).

Stearns, H. T., 1939, Geologic map and guide of the island of Oahu, Hawaii: Hawaii Div. Hydrography Bull. 2, 75 p.

—— 1940, Supplement to the geology and groundwater resources of the island of Oahu, Hawaii: Hawaii Div. Hydrography Bull. 5, 164 p.

—— 1966, Geology of the State of Hawaii: Palo Alto, California, Pacific Books, 266 p.

Stearns, H. T., and Vaksvik, K. N., 1935, Geology and ground-water resources of the island of Oahu, Hawaii: Hawaii Div. Hydrography Bull. 1, 478 p.

Strange, W. E., Machesky, L. F., and Woollard, G. P., 1965, A gravity survey of the island of Oahu, Hawaii: Pacific Sci., v. 19, p. 350–353.

Suhr, N. H., and Ingamells, C. O., 1966, Solution technique for analysis of silicates: Anal. Chemistry, v. 38, p. 730–734.

Swanson, D. A., 1972, Magma supply rate at Kilauea Volcano, 1952–1971: Science, v. 175, p. 169–170.

Tarling, D. H., 1962, Tentative correlation of Samoan and Hawaiian Islands using 'reversals' of magnetization: Nature, v. 196, p. 882–883.

—— 1965, The paleomagnetism of some of the Hawaiian Islands: Royal Astron. Soc. Geophys. Jour., v. 10, p. 93–104.

Watson, G. S., and Beran, R. J., 1967, Testing a sequence of unit vectors for serial correlation: Jour. Geophys. Research, v. 72, p. 5655–5659.

Watson, G. S., and Irving, E., 1957, Statistical

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

methods in rock magnetism: Royal Astron. Soc. Monthly Notices, v. 7, p. 289–300.

Wilson, J. T., 1963, A possible origin of the Hawaiian Islands: Canadian Jour. Physics, v. 41, p. 863–870.

Wilson, R. L., 1970, Permanent aspects of the earth's non-dipole magnetic field over upper Tertiary times: Royal Astron. Soc. Geophys. Jour., v. 19, p. 417–437.

Winchell, H., 1947, Honolulu series, Oahu, Hawaii: Geol. Soc. America Bull., v. 58, p. 1–48.

Manuscript Received by the Society June 2, 1972

Revised Manuscript Received August 30, 1972

Publication Authorized by the Director, U.S. Geological Survey